DeCovarT and CardamomOT: gene selection, noise layers and what a network means

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Abstract

DeCovarT (Chassagnol et al. 2023) estimates cell-type fractions from bulk RNA sequencing (RNA-seq), and models each type by a mean and a gene–gene covariance. CardamomOT (Maugé and Ventre 2026) infers a signed, directed gene regulatory network (GRN) from time-stamped single-cell snapshots, through a mechanistic bursting model. Both pick a small gene panel from a whole transcriptome, and both estimate a gene–gene structure, but for opposite reasons: in DeCovarT the covariance is a nuisance that weights genes, in CardamomOT the network is the estimand. We compare the two along five axes: (i) gene selection, (ii) expression versus measurement model, (iii) a speculative side axis (convolution, Wiener filter, mean-field ansatz), (iv) the diagonal of a network and self-regulation, (v) how many genes the data can support. On axis (v), DeCovarT needs a covariance, which shrinkage provides from two cells per type; reading a network requires the support of the precision, for which the number of cells per type must grow with the square of the largest number of neighbours of a gene. The code of CardamomOT 2.0.0 (repository state of 26 October 2026) adds a per-cell depth layer that the paper does not describe, and its diagnostic is the main item DeCovarT could borrow. CardamomOT could borrow a held-out likelihood for its penalty and a within-basin covariance. An appendix sketches a purely statistical counterpart: a mixture of dynamic Gaussian Bayesian networks, in which time carries self-regulation and feedback loops.

1 Notation, definitions and abbreviations

Sources. CardamomOT is read from its paper (PDF pages cited as 'p.') and from CardamomOT 2.0.0 (repository state of 26 October 2026).[1] All PDFs were read with local tools only. DeCovarT's symbols follow the notation section of the DeCovarT article. A bulk sample i is fixed inside a derivation, and its index returns in model statements. Table 1 lists the symbols, the Definitions box gives the technical terms with their formulas, and the abbreviations follow. A term or an abbreviation links to its entry at first use.

Table 1: Symbols. 'Article' is the DeCovarT article; 'paper' and 'code' are CardamomOT's.

Symbol	Meaning
Indices and counts	
$g, h = 1, \dots, G$	genes of the panel (signature); G is the number of genes kept for estimation. The CardamomOT paper writes n for this number (p. 27).
$j, \ell = 1, \dots, J$	cell types.
c, z_c	a reference cell, and its label: a cell type in DeCovarT, a promoter basin in CardamomOT.

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*INSERM. First draft, written with Claude Code (Anthropic). A grey bracketed identifier (V1a, V2b, …) points to a SymPy check in the ledger of Appendix B; a probabilistic statement is verified only under the assumptions stated with it.

[1] `https://github.com/eliasventre/CardamomOT`. Files read: `estimate_cell_depth.py`, `inference/depth.py`, `inference/depth_model.py`, `inference/gene_selection.py`, `inference/gandrillon.py`, `inference/network.py`. The repository ships no `CITATION.cff`.

Symbol	Meaning
C_j	number of reference cells of type j. Their profiles, conditional on the type, are the i.i.d. sample from which $\boldsymbol{\mu}_j$, $\boldsymbol{\Sigma}_{W,j}$ and $\boldsymbol{\Omega}_j$ are estimated. Each type is estimated on its own, with no structure shared across types; joint estimation is discussed in Section 3.4.
t	a time point (CardamomOT's snapshots).
i, n_i	a bulk sample, and its number of cells.
m_{ji}, $\boldsymbol{p}_{\cdot i}$	number of cells of type j in sample i, and the cell fractions $p_{ji} = m_{ji}/n_i$; $\boldsymbol{p}$ when i is fixed, $p = p_1$ when $J = 2$.
Counts and noise layers	
x_{gc}, $\boldsymbol{x}_{\cdot c}$	unique molecular identifier (UMI) count of gene g in cell c, and the vector over genes.
s_c	per-cell depth (size) factor: the measurement layer. Fang & Pachter's cell-wide factor $c_{\text{tech}}c_{\text{bio}}$ plays this role.
λ_{gc}; $\boldsymbol{\mu}_j^{\lambda}$, $\boldsymbol{\Sigma}_j^{\lambda}$	expression level of gene g in cell c, and its mean and covariance within type j: the expression layer.
$\boldsymbol{y}_{\cdot i}$	bulk expression vector of sample i.
DeCovarT moments and scales	
$\boldsymbol{\mu}_j$; $\boldsymbol{\mu}$	mean profile of type j on the linear UMI scale: per cell in the single-cell model (the article's $\tilde{\boldsymbol{\mu}}_{\cdot j}$), per population profile in the population model; $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_J)$ is $G \times J$.
$\boldsymbol{\Delta}$, Δ_g; σ_g^2	$\boldsymbol{\Delta} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ when $J = 2$, and its entry for gene g; a within-type variance of gene g common to the two types.
$\boldsymbol{\Sigma}_{W,j}$; $\boldsymbol{\Sigma}_W$	within-type cell-to-cell covariance of type j on the linear scale, from the cells of one specimen; $\boldsymbol{\Sigma}_W$ when it is common to all types.
$\boldsymbol{\Sigma}_j$	covariance of the type-j population profile (population references).
$\boldsymbol{\Omega}_j = (\omega_{gh})$	precision matrix $\boldsymbol{\Sigma}_{W,j}^{-1}$; ω_{gg} is the conditional precision of gene g (Definitions box).
$\boldsymbol{V}_j$; $\boldsymbol{V}$	empirical covariance of the C_j cells of type j, of rank at most $C_j - 1$; without an index, the marginal covariance of the mixed model of Appendix A.
S_j	mean transcriptome size of a type-j cell, $\mathbf{1}^{\mathsf{T}}\boldsymbol{\mu}_j$.
a_i; c_i, ϕ_i	sample-wide scale between the reference and the bulk library; the bulk mean and variance scales $c_i = a_i n_i$ and $\phi_i = a_i^2 n_i$. A sum of n_i independent cells, each scaled by a_i, has mean $a_i \sum_j m_{ji}\boldsymbol{\mu}_j = c_i \sum_j p_{ji}\boldsymbol{\mu}_j$ and covariance $a_i^2 \sum_j m_{ji}\boldsymbol{\Sigma}_{W,j} = \phi_i \sum_j p_{ji}\boldsymbol{\Sigma}_{W,j}$.
$\boldsymbol{D}$	diagonal bulk measurement noise (a nugget), absent from DeCovarT's model.
$\varphi_{\boldsymbol{z}}(\boldsymbol{t})$	characteristic function $\mathbb{E}\exp(\mathrm{i}\,\boldsymbol{t}^{\mathsf{T}}\boldsymbol{z})$ of a random vector $\boldsymbol{z}$: φ_j for one type-j cell, φ_j^{pop} for the type-j population profile, $\varphi_{\text{mix}} = \sum_j p_j\varphi_j$ for one cell drawn from the type mixture. Not the bulk scale ϕ_i.
Networks	
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	conditional-independence graph on the G genes: $gh \in \mathcal{E}$ if and only if $\omega_{gh} \neq 0$, $g \neq h$.
$\deg(g)$; $\deg_{\max}$	degree of gene g, $\deg(g) = \lvert\{h \neq g : \omega_{gh} \neq 0\}\rvert$, its number of neighbours; the maximum degree $\deg_{\max} = \max_g \deg(g)$.
$\mathrm{cl}(\mathcal{G})$, $\mathrm{tw}(\mathcal{G})$, $\mathrm{mlt}(\mathcal{G})$	clique number, treewidth, and maximum-likelihood threshold of $\mathcal{G}$ (Section 3.3).
λ, $\boldsymbol{\Lambda}$; λ_1, λ_2	graphical-lasso penalty and penalty matrix; the two penalties of the joint graphical lasso (Section 3.4). Not the expression level λ_{gc}.
$k = 1, \ldots, K$; C_k, $\boldsymbol{V}^{(k)}$, $\boldsymbol{\Omega}^{(k)}$	classes of a joint estimation (cell types, time points or batches), with their numbers of cells, empirical covariances and precision matrices.
CardamomOT	
$\boldsymbol{\theta} = (\theta_{gh})$; θ_{gg}	interaction matrix, the influence of gene h on gene g (p. 6); its diagonal, directed self-regulation.
$\boldsymbol{\theta}_0$	prior network towards which $\boldsymbol{\theta}$ is penalised (Eq. 10, p. 9).
β_g	basal parameter of gene g (the paper's β_i, p. 28).
B	number of promoter basins in the mixture of Eq. 4 (p. 7).
N_c	number of cells in a CardamomOT fit (p. 9); the paper's N_g is our G.

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Symbol	Meaning
R; R_1, R_0; α	candidate regulators of one target gene; true and null ones, $R = R_1 + R_0$; level of the quantile universal threshold (Section 3.5).
$\mathrm{NB}(\kappa, r)$	negative binomial (NB) law with shape κ and rate r, mean κ/r. The paper writes $\mathrm{NB}(a, b)$, the code $\mathrm{NB}(k, c)$; that k is not a class and that c is not a cell.
Linearised dynamics (Section 5)	
$\boldsymbol{A}$; $a_g = \gamma_g - \varsigma_g$	drift matrix; net decay of gene g, its degradation rate γ_g minus its linearised self-activation ς_g. Not the platform scale a_i.
b, c; $\boldsymbol{Q} = \mathrm{diag}(q_g)$	strengths of the edges $1 \to 2$ and $1 \to 3$; noise intensities.

Definitions

Best linear unbiased predictor In $\boldsymbol{y} = \boldsymbol{m} + \boldsymbol{Z}\boldsymbol{u} + \boldsymbol{e}$, with random effects $\boldsymbol{u} \sim (\boldsymbol{0}, \boldsymbol{\Gamma})$ independent of $\boldsymbol{e} \sim (\boldsymbol{0}, \boldsymbol{D})$ and $\boldsymbol{V} = \boldsymbol{Z}\boldsymbol{\Gamma}\boldsymbol{Z}^\mathsf{T} + \boldsymbol{D}$, the predictor of $\boldsymbol{u}$ that is linear in $\boldsymbol{y}$, unbiased, and of smallest mean squared error: $\hat{\boldsymbol{u}} = \boldsymbol{\Gamma}\boldsymbol{Z}^\mathsf{T}\boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{m})$, with the fixed part $\boldsymbol{m}$ replaced by its GLS estimate when it is unknown (Appendix A).

Shannon entropy, delta-entropy For a count X, $\mathcal{H}(X) = -\sum_{k \geq 0} \Pr(X = k) \log \Pr(X = k)$. Delta-entropy is a change of $\mathcal{H}$ for one gene between consecutive time points; the CardamomOT code uses $|\mathcal{H}(X_{g,t+1}) - \mathcal{H}(X_{g,t})|$, with $\mathcal{H}$ estimated by the BUB estimator.

Fisher information for $\boldsymbol{p}$ $\boldsymbol{I}(\boldsymbol{p}) = \mathbb{E}\{-\partial^2 \ell(\boldsymbol{p}) / \partial \boldsymbol{p}\, \partial \boldsymbol{p}^\mathsf{T}\}$, on the $J - 1$ free coordinates of the simplex. For $\boldsymbol{y} \sim \mathcal{N}\{\boldsymbol{m}(\boldsymbol{p}), \boldsymbol{\Sigma}(\boldsymbol{p})\}$, $I_{kl} = \partial_k \boldsymbol{m}^\mathsf{T} \boldsymbol{\Sigma}^{-1} \partial_l \boldsymbol{m} + \frac{1}{2} \mathrm{tr}(\boldsymbol{\Sigma}^{-1} \partial_k \boldsymbol{\Sigma}\, \boldsymbol{\Sigma}^{-1} \partial_l \boldsymbol{\Sigma})$. As a gene-selection objective, a panel $\mathcal{S}$ is scored by $\min_{\boldsymbol{p} \in \mathcal{P}} \log\det \boldsymbol{I}_{\mathcal{S}}(\boldsymbol{p})$ over a grid $\mathcal{P}$ of compositions (robust D-optimality, feature-selection vignette).

Hellinger distance $H(P, Q) \in [0, 1]$ with $H^2(P, Q) = 1 - \int \sqrt{f_P f_Q}$, for laws P and Q with densities f_P and f_Q.

Lyapunov equation For linear (Ornstein–Uhlenbeck) dynamics $\mathrm{d}\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}^*)\mathrm{d}t + \boldsymbol{B}\,\mathrm{d}\boldsymbol{W}$ with every eigenvalue of $\boldsymbol{A}$ of negative real part, the stationary covariance $\boldsymbol{\Sigma}$ solves $\boldsymbol{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\boldsymbol{A}^\mathsf{T} + \boldsymbol{Q} = \boldsymbol{0}$, with $\boldsymbol{Q} = \boldsymbol{B}\boldsymbol{B}^\mathsf{T}$.

Mahalanobis distance $\{(\boldsymbol{u} - \boldsymbol{v})^\mathsf{T} \boldsymbol{\Sigma}^{-1} (\boldsymbol{u} - \boldsymbol{v})\}^{1/2}$ between two points $\boldsymbol{u}$ and $\boldsymbol{v}$ for a covariance $\boldsymbol{\Sigma}$. Between two type means with a common $\boldsymbol{\Sigma}_W$, its square is $\boldsymbol{\Delta}^\mathsf{T} \boldsymbol{\Sigma}_W^{-1} \boldsymbol{\Delta}$.

Partial correlation, conditional precision For a Gaussian vector with precision $\boldsymbol{\Omega} = (\omega_{gh})$, $\rho_{gh|\mathrm{rest}} = -\omega_{gh} / (\omega_{gg}\omega_{hh})^{1/2}$ is the correlation of genes g and h given all other genes, and $\omega_{gg} = 1/\mathrm{Var}(x_g \mid \boldsymbol{x}_{-g})$ is the conditional precision of gene g. An edge of a GGM is a non-zero ω_{gh}, $g \neq h$.

Pearson residual $r_{gc} = (x_{gc} - \hat{\mu}_{gc}) / v(\hat{\mu}_{gc})^{1/2}$ for a fitted mean $\hat{\mu}_{gc}$ and a variance function v; for an NB law with shape κ_g, $v(\mu) = \mu + \mu^2/\kappa_g$.

Quasi-likelihood F test For a model with variance $\Phi_g v(\mu)$, two nested fits with deviances $\mathcal{D}_0 \geq \mathcal{D}_1$ that differ by q parameters give $F = \{(\mathcal{D}_0 - \mathcal{D}_1)/q\} / \hat{\Phi}_g$, compared with an $F_{q,\nu}$ law, ν the residual degrees of freedom. In the screen of Section 3, the null fit has one mean for all types and the alternative one mean per type, so $q = J - 1$; Lund et al. (2012) shrink $\hat{\Phi}_g$ across genes.

Wasserstein (Kantorovich) distance $W_r(P, Q) = \{\inf_\pi \mathbb{E}_\pi \|\boldsymbol{X} - \boldsymbol{Y}\|^r\}^{1/r}$, the infimum over couplings π, i.e. joint laws of $(\boldsymbol{X}, \boldsymbol{Y})$ with margins P and Q. On the real line, $W_1(P, Q) = \int |F_P(t) - F_Q(t)|\,\mathrm{d}t$ with F the distribution functions; this is the Kantorovich distance, and what `scipy.stats.wasserstein_distance` computes. For two Gaussian laws with a common covariance, $W_2 = \|\boldsymbol{\Delta}\|$.

Abbreviations

ANOVA analysis of variance.
BH Benjamini–Hochberg.
BLUP best linear unbiased predictor.
BUB best upper bound.
CLT central limit theorem.
CV coefficient of variation.
DAG directed acyclic graph.
DBN dynamic Bayesian network.
EM expectation–maximisation.
FC fold change.
FDR false discovery rate.
FFT fast Fourier transform.
FGL fused graphical lasso.
FPKM fragments per kilobase of transcript per million mapped reads.
FWER family-wise error rate.

GGL group graphical lasso.
GGM Gaussian graphical model.
GLMM generalised linear mixed model.
GLS generalised least squares.
GRN gene regulatory network.
JGCGM joint Gaussian copula graphical model.
LMM linear mixed model.
MAD median absolute deviation.
MLE maximum-likelihood estimate.
mlt maximum-likelihood threshold.
NB negative binomial.
W-NNLS weighted non-negative least squares.
OLS ordinary least squares.
PLN Poisson log-normal.
QUT quantile universal threshold.
REML restricted maximum likelihood.
RNA-seq RNA sequencing.
SCAD smoothly clipped absolute deviation.
scRNA-seq single-cell RNA sequencing.
SEM structural equation model.
TPM transcripts per million.
UMI unique molecular identifier.
WLS weighted least squares.

2 Aims, estimands and the two models side by side

Table 2: The two methods side by side. Page numbers refer to the CardamomOT paper unless a file name is given; 'code' means CardamomOT 2.0.0 (repository state of 26 October 2026).

	DeCovarT	CardamomOT
Data	bulk RNA-seq samples and one reference: sorted populations, or the cells of one specimen (Suppinger: single-cell GSE229513, bulk GSE229386, matched at 48 h)	scRNA-seq snapshots at successive time points, raw counts; optional prior network $\boldsymbol{\theta}_0$ and degradation rates (p. 8)
Estimand	fractions $\boldsymbol{p}_{\cdot i}$ of each bulk sample, with Wald intervals	signed directed $\boldsymbol{\theta}$, burst parameters, protein trajectories (pp. 6, 8)
Cell-level law	first two moments $(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_{W,j})$ per type on the linear UMI scale	$M_g \mid \boldsymbol{P} \sim$ NB from promoter bursting; a mixture over promoter basins, genes independent given the basin (Eq. 4, p. 7)
Aggregate	sum of n_i cells: $\boldsymbol{y}_{\cdot i} \sim \mathcal{N}\big(c_i \sum_j p_{ji}\boldsymbol{\mu}_j,\ \phi_i \sum_j p_{ji}\boldsymbol{\Sigma}_{W,j}\big)$, linear in $\boldsymbol{p}$; population references give $\sum_j p_{ji}^2 \boldsymbol{\Sigma}_j$, quadratic [V6a,V6c]	none (no bulk); the population marginal is the basin mixture
Noise layers	expression and measurement mixed in $\boldsymbol{\Sigma}_{W,j}$; bulk scales c_i, ϕ_i	paper: expression (bursting) only; code: per-cell depth factor s_c (Section 4)
Gene selection	detection, presence in every type, consistency screen, optional Fisher-information refinement (Section 3)	paper: core set plus databases (p. 28); code: entropy and Wasserstein terminals, basin separability, coarse network, Steiner closure
Role of the network	nuisance plug-in that weights genes (GLS)	the object of inference
Sparsity	dense shrinkage $\hat{\boldsymbol{\Sigma}}$ by default; graphical lasso with cross-validated λ as an alternative	ℓ_1 towards $\boldsymbol{\theta}_0$ with $\lambda = 0.05 N_c/\sqrt{G}$ (Eq. 10, p. 9); code: quantile universal threshold at level 0.5 per target gene, then an adaptive-lasso refinement (`network.py`; Section 3.5)
Diagonal	$\omega_{gg} = 1/\operatorname{Var}(x_g \mid \boldsymbol{x}_{-g})$, never penalised	θ_{gg}, directed self-regulation; 'no specific penalty' (p. 28)
Inference	maximum likelihood, profile likelihood, Wald intervals for $\boldsymbol{p}$	point estimates; run-to-run variability over 3–5 seeds (p. 27)

The two models answer different questions (Table 2). DeCovarT is a multivariate regression in the strict sense: several correlated responses (the genes) share one design (the fractions). Its gene covariance matters because it weights genes, as in generalised least squares (GLS). CardamomOT is a mechanistic model of the cell, and its network is what the user reads. A good covariance for DeCovarT need not be a good network for CardamomOT, and the converse also holds (Section 5).

3 Derivation of regularised covariance or regulatory networks(axes i and v)

3.1 CardamomOT

The paper treats the panel as prior information. 'All experimental applications presented here operate on gene subsets rather than the full transcriptome (Appendix C), which is not a neutral choice'

(p. 27); 'the selected gene set should be recognised as a form of prior information in its own right' (p. 27). It recommends a core set of nodes from prior knowledge, completed by curated interaction databases and a coarse genome-wide network, and calls the automation 'ongoing work' (p. 28). For Semrau and Kameneva, the applications use gene lists from earlier work; for Schiebinger, markers plus 'dynamically variable genes' ranked by entropy and Wasserstein distances (p. 32).

The code of CardamomOT 2.0.0 automates that recommendation (module docstring of `inference/gene_selection.py`):

- **gene universe:** mitochondrial, ribosomal-protein and non-coding genes are excluded by name or biotype;
- **terminal genes** (`inference/gandrillon.py`), from Gandrillon's workflow. 'KD' is the one-dimensional Wasserstein distance between the counts of a gene at two consecutive time points, divided by the gene's standard deviation over the pair. 'MDE' is the absolute change of a discrete Shannon entropy estimated by the best upper bound (BUB) estimator (Paninski 2003). Terminals are then chosen by a round robin over cell types, each tested against the rest by a Wilcoxon test on log-normalised counts (Scanpy's `rank_genes_groups`) (Wolf et al. 2018);
- **basin separability** (`nb_init_separation`): genes whose negative-binomial modes are well separated;
- **a coarse network** on highly variable genes (OTVelo-Corr);
- **closure:** a budgeted prize-collecting Steiner selection that anchors the terminals to the stimuli.

Gandrillon's criteria. The 'Kantorovich distance' cited by Gandrillon (p. 7, to Vershik 2006) is the Wasserstein-1 distance W_1, and the code computes it with `scipy.stats.wasserstein_distance`.

3.2 DeCovarT

The estimand is a gene panel that makes $(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_{W,j})$ estimable and informative about $\boldsymbol{p}$, not population-level differential expression. With one specimen, cells are not replicates, so a p-value describes a screen inside that specimen (Zimmerman et al. 2021; Squair et al. 2021). The tiers are written in the feature-selection vignette:

- **detection:** genes detected in too few reference cells are removed;
- **presence in every type**, so that no within-type variance V_{gg} is zero: necessary for a non-singular estimate of $\boldsymbol{\Sigma}_{W,j}$, not sufficient (Section 3.3); this also removes perfect markers, for which proportions are not identified once gene-wise platform ratios are allowed (Xie et al. 2026, p. 7);
- **consistency screen** ('marker gene consistency', Wang et al. (2019, p. 2)): the between/within ratio on Pearson residuals, or its count-scale versions, the analysis-of-deviance and quasi-likelihood F tests (Lund et al. 2012); the test is a gate, and genes are ranked by effect size;
- **optional refinement** by the Fisher information for $\boldsymbol{p}$, which needs $\boldsymbol{\Sigma}_{W,j}$.

For two types and one gene, under the single-cell model with equal within-type variances σ_g^2, the Fisher information for p is $n_i\Delta_g^2/\sigma_g^2$, with $\Delta_g = \mu_{g1} - \mu_{g2}$ (Appendix A) [V4a]. The analysis of variance (ANOVA) ratio on $C_1 = C_2 = C$ cells per type is the plug-in estimate of $(C/2)\Delta_g^2/\sigma_g^2$, so both rank genes alike; its mean is close to $1 + (C/2)\Delta_g^2/\sigma_g^2$ [V4b]. With unequal variances or correlated genes they do not: the multivariate information with a common $\boldsymbol{\Sigma}_W$ is $n_i\boldsymbol{\Delta}^\mathsf{T}\boldsymbol{\Sigma}_W^{-1}\boldsymbol{\Delta}$, a squared Mahalanobis distance [V4d]. With a common $\boldsymbol{\Sigma}_W$, the squared Hellinger distance between the two type laws is $1 - \exp(-\frac{1}{8}\boldsymbol{\Delta}^\mathsf{T}\boldsymbol{\Sigma}_W^{-1}\boldsymbol{\Delta})$, increasing in that distance [V8a]; the 2-Wasserstein distance is not, since for equal covariances it reduces to $\|\boldsymbol{\Delta}\|$ [V8b], and the two can rank candidate panels in opposite orders [V8c]. This links the screen to the overlap criteria of the screening scenario. The one-gene view is the screen, not the objective: with type-specific covariances, both the Fisher information for p and the overlap of the two Gaussian type laws add a covariance-difference term to the Mahalanobis separation of the means (Appendix A, Equations (10) and (11)).

The two methods push the within-group variance in opposite directions. CardamomOT keeps genes whose distribution changes in time and splits into separable basins, so multimodality is wanted.

DeCovarT wants a unimodal law within each type, with a covariance it can estimate.

3.3 How many genes can the data support?

The answer depends on what is estimated from the C_j cells of each type. The box below separates the two goals; the symbols are those of Table 1.

Identifiability: covariance versus network

(a) The covariance $\boldsymbol{\Sigma}_{W,j}$: DeCovarT's plug-in. $\boldsymbol{\Sigma}_{W,j}$ holds every association between two genes, direct or through other genes. DeCovarT's likelihood needs a positive-definite estimate close to $\boldsymbol{\Sigma}_{W,j}$ in a covariance loss, such as the expected Gaussian log-score of Section 7 or a Frobenius norm. Whether the zeros of $\hat{\boldsymbol{\Omega}}_j$ are the true ones does not enter. What matters is that the estimate is well defined (finite, positive definite and unique), not that some algorithm returns a matrix.

- **The unconstrained maximum-likelihood estimate (MLE).** With the mean estimated, it is the empirical covariance $\boldsymbol{V}_j = \boldsymbol{X}_c^\mathsf{T}\boldsymbol{X}_c/C_j$, where $\boldsymbol{X}_c$ is the centred $C_j \times G$ matrix of the type-j cells. Its rank is that of $\boldsymbol{X}_c$, at most $\min(C_j - 1, G)$, with equality for data in general position, which holds almost surely for Gaussian cells: $C_j\boldsymbol{V}_j \sim W_G(C_j - 1, \boldsymbol{\Sigma}_{W,j})$ (Muirhead 1982, ch. 3) [V10a]. $\boldsymbol{V}_j$ is positive definite if and only if $\operatorname{rank}\boldsymbol{X}_c = G$. This needs $C_j \geq G+1$ *and* no exact linear or affine relation between genes; such a relation lowers the rank further [V10b], and with $C_j \leq G$ the centred profiles always satisfy one. When $\boldsymbol{V}_j$ is singular, the Gaussian log-likelihood is unbounded above along its null directions, and the MLE does not exist [V10c].
- **The presence gate is necessary, not sufficient.** A gene constant within type j has $V_{gg} = 0$, the simplest exact relation; the gate of Section 3.2 removes it, and it keeps the diagonal shrinkage target of Table 3 positive definite. With $C_j \leq G$, or another exact relation, $\boldsymbol{V}_j$ stays singular after the gate [V10b].
- **Relaxing the requirement** (Table 3). The penalty, the constraint or the structure, not the data, makes the estimate exist when $C_j \leq G$.

Estimator	Formula	Unique positive-definite estimate when
Unconstrained MLE	$\boldsymbol{V}_j = \boldsymbol{X}_c^\mathsf{T}\boldsymbol{X}_c/C_j$	$C_j \geq G+1$ and $\operatorname{rank}\boldsymbol{X}_c = G$ [V10a,V10b,V10c]
Linear shrinkage	$(1-\rho)\boldsymbol{V}_j + \rho\boldsymbol{T}$, $\rho \in (0,1]$; Ledoit–Wolf $\boldsymbol{T} = (\operatorname{tr}\boldsymbol{V}_j/G)\boldsymbol{I}$, Schäfer–Strimmer $\boldsymbol{T} = \operatorname{diag}(\boldsymbol{V}_j)$	$C_j \geq 2$, $\boldsymbol{T} \succ 0$ and $\hat{\rho} > 0$ [V11a,V11b,V11c]
Condition-number constraint	$\max \ell(\boldsymbol{\Sigma})$ subject to $\lambda_{\max}(\boldsymbol{\Sigma})/\lambda_{\min}(\boldsymbol{\Sigma}) \leq \kappa_{\max}$; eigenvalues of $\boldsymbol{V}_j$ truncated to $[\tau^\star, \kappa_{\max}\tau^\star]$	$C_j \geq 2$, $\boldsymbol{V}_j \neq \boldsymbol{0}$ (our reading for an estimated mean)
Structured $\boldsymbol{\Sigma}$	factor $\boldsymbol{B}\boldsymbol{B}^\mathsf{T} + \boldsymbol{\Psi}$ ($\boldsymbol{B}$ of size $G \times r$, $\boldsymbol{\Psi}$ diagonal), banded, or linear $\boldsymbol{G}_0 + \sum_k v_k\boldsymbol{G}_k$	$\operatorname{rank}\boldsymbol{V}_j = G$ in the result we read; below that, to be shown per structure
Graphical lasso	$\max_{\boldsymbol{\Omega}\succ 0} \log\det\boldsymbol{\Omega} - \operatorname{tr}(\boldsymbol{V}_j\boldsymbol{\Omega}) - \lambda\sum_{g\neq h}\lvert\omega_{gh}\rvert$	$C_j \geq 2$, $V_{gg} > 0$, $\lambda > 0$

Table 3: When each covariance estimator of type j is well defined. ℓ is the Gaussian log-likelihood, $\lambda_{\max}$ and $\lambda_{\min}$ the extreme eigenvalues.

- **Shrinkage** (Schäfer and Strimmer 2005) (Ledoit and Wolf 2004)[a]: $\lambda_{\min}\{(1-\rho)\boldsymbol{V}_j + \rho\boldsymbol{T}\} \geq \rho\,\lambda_{\min}(\boldsymbol{T})$, whatever the rank of $\boldsymbol{V}_j$ [V11b]. The estimated intensity $\hat{\rho}$ is truncated to $[0,1]$, and $\hat{\rho} = 0$ returns $\boldsymbol{V}_j$. The diagonal target needs every $V_{gg} > 0$, hence the presence gate; the scaled identity needs only $\boldsymbol{V}_j \neq \boldsymbol{0}$.
- **Condition number** (Won et al. 2013): they maximise the Gaussian log-likelihood under $\operatorname{cond}(\boldsymbol{\Sigma}) \leq \kappa_{\max}$ (eq. 6, p. 4). The solution keeps the eigenvectors of $\boldsymbol{V}_j$ and replaces each eigenvalue l_g by $\min\{\max(\tau^\star, l_g), \kappa_{\max}\tau^\star\}$ (eq. 7, p. 5), with $\tau^\star$ in closed form and found in $O(G)$ operations (eqs 11–12, p. 7). The solution is unique when $\kappa_{\max} \leq \operatorname{cond}(\boldsymbol{V}_j)$ (Theorem 1, pp. 6–7), which always holds for a singular $\boldsymbol{V}_j$, and it is well conditioned 'even when $n < p$' (p. 5). $\kappa_{\max}$ is chosen by predictive likelihood (Section 3, p. 9). Their model has a known zero mean; with an estimated mean we read the condition as $\boldsymbol{V}_j \neq \boldsymbol{0}$, i.e. $C_j \geq 2$ with one non-constant gene. No sparsity is assumed.
- **Structure on $\boldsymbol{\Sigma}$.** A factor, low-rank-plus-diagonal, banded or linear model has fewer parameters than $G(G+1)/2$. (Zwiernik et al. 2017) study the linear models $\boldsymbol{\Sigma}_v = \boldsymbol{G}_0 + \sum_k v_k\boldsymbol{G}_k$, which include correlation matrices and prescribed zeros in $\boldsymbol{\Sigma}$ (pp. 1269–1270). The likelihood is not concave there; it is strictly concave on $\{\boldsymbol{\Sigma} : \boldsymbol{0} \prec \boldsymbol{\Sigma} \prec 2\boldsymbol{V}_j\}$ (their Proposition 1, p. 1273), and for $C_j \gtrsim 14G$ hill climbing from the least-squares estimate reaches the global maximum with high probability (p. 1269). Their

existence result assumes a full-rank sample covariance (Proposition 5, p. 1278), so it does not relax $C_j \geq G + 1$; their 'with probability 1 when $n \geqslant p$' should read $n \geq p + 1$ when the mean is estimated [V10d]. Below full rank, existence depends on the structure; for a factor model a variance $\psi_g \to 0$ (a Heywood case) can make the likelihood unbounded (our reading, not checked).

- **Graphical lasso**: with $\lambda > 0$, positive V_{gg} and an unpenalised diagonal, it has a unique positive-definite solution for any $C_j \geq 2$, even when $C_j < G$ (Carter 2025). The penalty, not the data, makes the estimate exist below $C_j = G + 1$.

None of these estimators needs a sparse truth.

(b) The support of $\boldsymbol{\Omega}_j$: the network. This is CardamomOT's kind of estimand: its $\boldsymbol{\theta}$ is directed, but it is read through its non-zero entries. On the graph $\mathcal{G}$ of $\boldsymbol{\Omega}_j$:

- a *clique* is a set of genes pairwise joined by edges; the clique number $\mathrm{cl}(\mathcal{G})$ is the size of the largest clique;
- a graph is *chordal* if every cycle through four or more genes has a chord, an edge between two genes that are not consecutive on the cycle. The treewidth plus one, $\mathrm{tw}(\mathcal{G}) + 1$, is 'the clique number of a minimal chordal cover', the smallest clique number of a chordal graph that contains $\mathcal{G}$ (Bernstein and Outlaw 2023) (p. 2);
- the degree $\deg(g)$ counts the neighbours of gene g, and $\deg_{\max}$ is its maximum over genes.

The requirements, from the weakest to the strongest:

1. **Existence** of the penalised estimate, as in (a). The graphical lasso estimator 'always exists' (Bernstein & Outlaw, p. 2); its convergence says nothing about the graph.
2. **Existence of the MLE on a fixed graph**, i.e. of the refit under the selected zeros. For a Gaussian model $\mathcal{M}$ with mean zero, 'the minimum n such that the MLE exists for almost every $X \in \mathbb{R}^{p \times n}$ is called the maximum likelihood threshold (MLT) of $\mathcal{M}$', and the model of a graph with edge set E is $\mathcal{M}_{\mathcal{G}} = \{K \succ 0 : K_{ij} = 0 \text{ whenever } ij \notin E\}$ (Bernstein & Outlaw, p. 2); their n and p are our C_j and G, and we write $\mathrm{mlt}(\mathcal{G})$ for the maximum-likelihood threshold (mlt). They state these bounds:
   - $\mathrm{mlt}(\mathcal{G}) \leq G$, from Dempster (p. 2);
   - $\mathrm{cl}(\mathcal{G}) \leq \mathrm{mlt}(\mathcal{G}) \leq \mathrm{tw}(\mathcal{G}) + 1$, from Buhl (p. 2); a chordal graph is its own minimal chordal cover, so mlt = cl for chordal graphs (Uhler 2012; Gross and Sullivant 2018);
   - $\mathrm{mlt}(\mathcal{G})$ is at most the generic completion rank of $\mathcal{G}$, with equality for nine or fewer vertices (their Theorem 2.2, p. 3);
   - the k-core of $\mathcal{G}$ is what remains after repeatedly deleting genes of degree less than k, and the smallest k with an empty k-core bounds $\mathrm{mlt}(\mathcal{G})$ from above (p. 5, after Gross & Sullivant); sparse graphs have small bounds.

   Their model has a known zero mean. With the mean estimated from the C_j cells, the centred sample has rank $C_j - 1$, so the requirement reads $C_j - 1 \geq \mathrm{mlt}(\mathcal{G})$ (our reading). In their simulations (at most nine variables, independent Gaussian data), the probability that the graph selected by the graphical lasso has an mlt within the sample size rises with the penalty and with the sample size; with one observation, the graphical lasso 'never selected the graphical model with no edges, the only model with MLT 1' (pp. 4–5).
3. **Support recovery**: $C_j \gtrsim \deg_{\max}^2 \log G$ for the graphical lasso, against $C_j \gtrsim \deg_{\max} \log G$ for neighbourhood selection (Chiquet 2015, p. 41), under incoherence (irrepresentability), bounded eigenvalues and a minimum $|\omega_{gh}|$ (Ravikumar et al. 2011). These conditions involve the unknown $\boldsymbol{\Sigma}_{W,j}$, and 'we can never verify them on real data' (Chiquet 2015, p. 77). With $G = 500$ and $\deg_{\max} = 10$, $\deg_{\max}^2 \log G \approx 620$ before constants, and hubs raise $\deg_{\max}$.

DeCovarT's plug-in needs (a) only. Reading $\hat{\boldsymbol{\Omega}}_j$, or CardamomOT's $\hat{\boldsymbol{\theta}}$, as a network needs (b).

[a] Overview: Wikipedia, 'Estimation of covariance matrices', shrinkage estimation.

CardamomOT has G^2 interaction parameters ('10^4 for $n = 100$', p. 27, where n is our G) and recommends 'not exceeding of the order of 100 genes in the fully unconstrained setting' (p. 28), with more genes allowed when $\boldsymbol{\theta}_0$ fixes more entries. DeCovarT faces the same trade-off per type: several Suppinger types have $C_j \leq G$, which rules out the unpenalised covariance and puts support recovery out of reach for all but very sparse graphs. Its plug-in only needs goal (a), which is why dense shrinkage (Schäfer and Strimmer 2005) is the default.

Selection removes what it should estimate. A panel chosen for small within-type variance, such

as ReDeconv's smallest coefficients of variation (CVs) (Lu et al. 2025, p. 14) or MuSiC's 'cross-cell consistent genes' (Wang et al. 2019, p. 3), leaves covariances dominated by Poisson sampling. Their correlations are then attenuated (Section 4), and the estimated precision looks diagonal for reasons that have nothing to do with regulation.

3.4 Penalties across gene pairs and across classes

Prior structure. An edge-weighted graphical lasso takes a penalty matrix $\mathbf{\Lambda} = (\lambda_{gh})$: small weights for whitelisted pairs, zero constraints for blacklisted ones (Friedman et al. 2008). CardamomOT's prior network, or an OmniPath prior as in its repository (p. 27), can enter DeCovarT this way. Neighbourhood selection fits G lasso regressions (Meinshausen and Bühlmann 2006); it scales well but returns a graph, not a positive-definite $\hat{\mathbf{\Sigma}}$, so a refit under the selected zeros is needed, which is requirement (b)2 of Section 3.3. A penalty chosen for prediction does not select the true neighbourhood (their Proposition 1, p. 5), which matters for CardamomOT, not for DeCovarT's plug-in.

Several classes at once. DeCovarT estimates each $\mathbf{\Sigma}_{W,j}$ from the cells of type j alone. Joint estimation couples K precision matrices $\mathbf{\Omega}^{(1)}, \ldots, \mathbf{\Omega}^{(K)}$, whose classes can be cell types, time points or batches. The methods differ in what the classes share.

- **The support only.** The graphical group lasso of Chiquet et al. (2011) groups the K partial correlations of each gene pair, so that edges are 'present or absent together across the conditions', 'not necessarily with the same strength'; an edge present in one class and absent in another then has probability zero (Chiquet 2015, p. 51). The cooperative lasso splits each group into its positive and negative parts: an edge may vanish in some classes, but it keeps one sign where it is present, since 'sign swaps are very unlikely' (p. 51). The group graphical lasso (GGL) in the box below is of the same kind.
- **The weights as well.** The fused graphical lasso (FGL) (Danaher et al. 2012) also penalises differences between edge values (box below). PLNFGL (Zhai et al. 2026) estimates each class covariance under a Poisson log-normal (PLN) model by moments and embeds it into the fused graphical lasso (their eq. 6, p. 3); its authors note that this step 'may face computational challenges when applied to large gene sets or to datasets involving more than five conditions' (p. 10). The joint Gaussian copula graphical model (JGCGM) (Wu et al. 2020) assumes a nonparanormal law (a Gaussian copula) in each class and writes each precision as a common part plus a class-specific part, $\mathbf{\Omega}^{(k)} = \boldsymbol{M} + \boldsymbol{H}^{(k)}$, with ℓ_1 penalties weighted by aK on $\boldsymbol{M}$ and by $1-a$ on each $\boldsymbol{H}^{(k)}$, $0 < a < 1$ (p. 4).
- **The difference only.** For two classes, the lasso-penalised D-trace loss (Yuan et al. 2017) estimates $\mathbf{\Omega}^{(2)} - \mathbf{\Omega}^{(1)}$ 'without attempting to estimate precision matrices' (p. 2): the loss has its unique minimiser at the difference of the two inverse covariances (p. 5). Only the difference needs to be sparse.

These penalties serve goal (b). For DeCovarT's plug-in, a joint penalty helps only if it lowers the covariance loss of every type, and the weighting caveat in the box below makes that doubtful for rare types (our reading, not tested).

Joint graphical lasso: group versus fused penalty

Danaher et al. (2012) estimate K precision matrices from classes of C_k cells with empirical covariances $\boldsymbol{V}^{(k)}$. In our notation, they maximise (their eq. 2.4, p. 3)

$$\sum_{k=1}^{K} C_k \left\{ \log\det \mathbf{\Omega}^{(k)} - \operatorname{tr}\left(\boldsymbol{V}^{(k)} \mathbf{\Omega}^{(k)}\right) \right\} - P\left(\mathbf{\Omega}^{(1)}, \ldots, \mathbf{\Omega}^{(K)}\right) \tag{1}$$

over positive-definite $\boldsymbol{\Omega}^{(1)}, \ldots, \boldsymbol{\Omega}^{(K)}$, with one of two penalties (eqs 2.5 and 2.6, p. 4):

$$P_{\mathrm{FGL}} = \lambda_1 \sum_{k=1}^{K} \sum_{g \neq h} |\omega_{gh}^{(k)}| + \lambda_2 \sum_{k<k'} \sum_{g,h} |\omega_{gh}^{(k)} - \omega_{gh}^{(k')}|, \tag{2}$$

$$P_{\mathrm{GGL}} = \lambda_1 \sum_{k=1}^{K} \sum_{g \neq h} |\omega_{gh}^{(k)}| + \lambda_2 \sum_{g \neq h} \Big\{ \sum_{k=1}^{K} \big(\omega_{gh}^{(k)}\big)^2 \Big\}^{1/2}. \tag{3}$$

- **Fused versus group.** A large λ_2 in (2) makes many entries identical across classes. With $\lambda_1 = 0$ and $\lambda_2 > 0$, (3) gives every $\hat{\boldsymbol{\Omega}}^{(k)}$ the same pattern of non-zero entries. 'GGL encourages a weaker form of similarity across the K precision matrices than does FGL: the latter encourages shared edge values across the K matrices, whereas the former encourages only a shared pattern of sparsity' (p. 4).
- **Classes.** Cell types, time points or batches. For one cell type followed over ordered time points $k = 1, \ldots, K$, the fused sum can be restricted to consecutive times, $\lambda_2 \sum_{k=1}^{K-1} \sum_{g,h} |\omega_{gh}^{(k+1)} - \omega_{gh}^{(k)}|$: the network may drift, but each change between neighbouring times is penalised. This is our adaptation; (2) fuses every pair of classes, and Danaher et al. point to time-varying network methods for ordered classes (p. 8).
- **Weighting caveat.** In (1), 'the contribution of each class to the penalized log likelihood is weighted by its size; consequently, the largest class can have outsize influence on the estimated networks' (p. 13); in their lung application they weighted the classes equally (p. 12). Chiquet (2015, p. 49) sets the same condition for one penalty shared by all classes: it is sensible 'provided that the C samples have similar sizes'. The JGCGM objective is weighted by the class sizes in the same way (eq. 2, p. 3). With Suppinger's types, the edges of a rare type with a small C_k would be fused towards, or grouped with, those of the abundant types, and edges specific to the rare type could be set to zero.

3.5 CardamomOT's ℓ_1 penalty: its level, an error-rate reading, alternatives

Paper. Step 2 of the inference updates the network by (Eq. 10, p. 9)

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{c} \mathrm{loss}\big\{k_{\mathrm{on}}^{\boldsymbol{\theta}}(\boldsymbol{P}_c), \boldsymbol{k}_{z_c}\big\} + \lambda \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|_1, \tag{4}$$

a cross-entropy between the predicted burst rates and the basin modes, summed over the cells c. The paper sets '$\lambda = 0.05 \cdot N_c/\sqrt{N_g}$, where N_c is the number of cells, reflecting the heuristic that the number of active regulators per gene scales at most as $\sqrt{N_g}$, and that a per-gene basin label attribution error of 5 % is a reasonable target for reliable inference' (p. 9); N_g is our G.

Code (version 2.0.0). This formula does not appear. `inference/network.py` fits one target gene g at a time, with one level λ_g for the row $\boldsymbol{\theta}_{g\cdot}$. The function `qut_penalty` returns the quantile universal threshold (QUT) (Giacobino et al. 2017): the `level`-quantile of $\max_h |\partial L/\partial \theta_{gh}|$ at the null fit (no interaction, per-sample basal parameters fitted), over permutations of the target within each sample. The pipeline passes `qut_level=0.5` and `qut_n_perm=20` (`model/base.py`), which override the function's own default level of 0.95. The adaptive-lasso refinement has its own threshold at the same level (`qut_garrote`). The ℓ_1 norm is smoothed, quadratic for $|x| < 10^{-3}$.

Where the two levels come from, and what they control (our reading)

- **The paper's λ as a loss budget.** For one target gene, $\sqrt{G}$ edges of unit size $|\theta_{gh} - \theta_{0,gh}| = 1$ cost $\lambda\sqrt{G} = 0.05 N_c$ [V12d]: the loss of mislabelling 5 % of the N_c cells if a wrong basin costs about one unit of cross-entropy. Edges enter only if they buy more than that. The paper gives no derivation; this is our reconstruction.
- **Its scale.** $\lambda/N_c = 0.05/\sqrt{G}$ is fixed, so λ is a floor on the mean per-cell score of an edge, not a significance level. The null score, a sum of N_c independent per-cell terms, has a standard deviation of order $\sqrt{N_c}$, so λ over the noise grows as $\sqrt{N_c}$ [V12d]: few false edges in large data sets, but weak true edges never enter.
- **The QUT.** By the Karush–Kuhn–Tucker conditions, the fit of target g has no edge if and only if $\lambda_g \geq \max_h |s_h|$, with $s_h = \partial L/\partial \theta_{gh}$ at the null fit. λ_{QUT} is the $(1-\alpha)$ quantile of $\max_h |s_h|$ under the null: the smallest λ that returns the empty model with probability $1-\alpha$. The code takes the null by

permutation and $\alpha = 1 - \texttt{qut_level} = 0.5$, estimated from 20 permutations.

Assumptions for the next items: target g has R candidate regulators, R_1 true and $R_0 = R - R_1$ null; the null scores are independent and Gaussian (orthogonal design) and do not depend on the true edges; the true scores lie far above the threshold. The ledger writes m for R and k for R_1.

- **family-wise error rate (FWER).** λ_{QUT} controls the probability of at least one false edge into g at α under the global null of g. With R_1 true regulators, the per-target FWER is $1 - (1-\alpha)^{R_0/R} \le \alpha$ [V12c]. The control is per target gene: over G independent fits, the network-wide FWER is close to 1 at $\alpha = 0.5$.
- **Bonferroni.** By the union bound, λ_{QUT} is never above the Bonferroni threshold $\sigma\Phi^{-1}\{1-\alpha/(2R)\}$, whatever the dependence between scores. Under independence it is the Šidák threshold, with per-test level $1-(1-\alpha)^{1/R} \ge \alpha/R$ [V12a]. Both thresholds grow as $\sigma\sqrt{2\log R}$ and their ratio tends to 1 [V12b]: at $R = 100$ it is 0.998 for $\alpha = 0.05$ and 0.962 for $\alpha = 0.5$. The per-test levels differ by the factor $-\log(1-\alpha)/\alpha$, 1.03 at $\alpha = 0.05$ and 1.39 at $\alpha = 0.5$ [V12a].
- **False edges and an FDR-type level.** The expected number of false edges into g tends to $-\log(1-\alpha)$ when $R_1 = \sqrt{R}$ and $R \to \infty$ [V12c]: 0.69 at $\alpha = 0.5$, about $0.7G$ false edges in the network. With $R = 100$ and $R_1 = 10$ (the paper's $\sqrt{G}$), the false-edge share $\mathbb{E}V/(\mathbb{E}V + R_1)$ is 0.059 at $\alpha = 0.5$ and 0.005 at $\alpha = 0.05$ [V12c]; a ratio of expectations, not the FDR itself. A share q needs the per-test level $t = qR_1/\{(1-q)R_0\}$, about $qR_1/\{(1-q)\alpha\}$ times the Bonferroni level α/R [V12c]: an FDR-type target is about R_1 times more permissive than a FWER one.
- **Dependence.** Regulator proteins are correlated, so the scores are not independent. The within-sample permutation keeps that dependence: the QUT still controls the per-target FWER under the permutation null, and stays below the Bonferroni threshold by the union bound. The expected counts and shares above assume independence. With 20 permutations the median itself is uncertain.

Alternatives to the element-wise ℓ_1 on $\boldsymbol{\theta}$. $\boldsymbol{\theta}$ is directed ($\theta_{gh} \ne \theta_{hg}$) and enters a logistic burst rate, so its likelihood is not a function of a covariance.

- **D-trace loss.** Yuan et al. (2017) minimise $L_D(\boldsymbol{\Delta}) = \frac{1}{4}(\langle \boldsymbol{\Sigma}_X\boldsymbol{\Delta}, \boldsymbol{\Delta}\boldsymbol{\Sigma}_Y\rangle + \langle \boldsymbol{\Sigma}_Y\boldsymbol{\Delta}, \boldsymbol{\Delta}\boldsymbol{\Sigma}_X\rangle) - \langle \boldsymbol{\Delta}, \boldsymbol{\Sigma}_X - \boldsymbol{\Sigma}_Y\rangle$, convex with unique minimiser $\boldsymbol{\Sigma}_Y^{-1} - \boldsymbol{\Sigma}_X^{-1}$ (eq. 1, p. 5), a generalisation of the D-trace loss of Zhang and Zou (2014) (p. 4). It needs a symmetric parameter and second-moment matrices as data; neither holds for $\boldsymbol{\theta}$. Two ideas carry over: a loss whose minimiser is the target without a log-determinant, and a penalty on a difference only. The code already does the second across conditions, $\boldsymbol{\theta}_c = \boldsymbol{\theta}_{\mathrm{shared}} + \boldsymbol{\delta}_c$ with an ℓ_1 on $\boldsymbol{\delta}_c$ (`core_inference_conditions`). A directed analogue of the D-trace loss would be a squared residual of the Lyapunov equation of Section 5, $\|\boldsymbol{A}\boldsymbol{V}_b + \boldsymbol{V}_b\boldsymbol{A}^{\mathsf{T}} + \boldsymbol{Q}\|_F^2$ per basin b, with an ℓ_1 on the off-diagonal of $\boldsymbol{A}$ (speculative: it rests on the linearisation).
- **Log-determinant loss** (graphical lasso). It needs a symmetric $\boldsymbol{\Omega} \succ 0$ and a Gaussian likelihood, so it does not apply to $\boldsymbol{\theta}$. Its existence argument does carry over: the penalty makes the estimate exist. An unpenalised logistic fit diverges for a separable target, and the code bounds those edges with a Gaussian prior (`ridge_coefficients`, 'bounds the edges of nearly separable targets'). The graphical lasso would fit the within-basin covariance that CardamomOT lacks (Section 7), as an undirected layer next to $\boldsymbol{\theta}$.
- **Group or cooperative lasso.** Chiquet's eqs 2.20–2.21 group the partial correlations of a gene pair across conditions; the cooperative version splits each group into positive and negative parts, so an edge may vanish in some classes but keeps one sign (Chiquet 2015, p. 51) (Chiquet et al. 2012). For $\boldsymbol{\theta}$ the classes can be conditions or cell types (the code fuses edge values; a group penalty would share only the support), time intervals (fused between consecutive times, as in the box of Section 3.4, or grouped), or the B basins of the target. With basin-specific interactions $\boldsymbol{\theta}_{gh} = (\theta_{gh}^{(1)}, \dots, \theta_{gh}^{(B)})$ and prior weights w_h, the cooperative penalty reads

$$P(\boldsymbol{\theta}_{g\cdot}) = \lambda \sum_h w_h \big\{ \|(\boldsymbol{\theta}_{gh} - \theta_{0,gh}\mathbf{1})_+\|_2 + \|(\boldsymbol{\theta}_{gh} - \theta_{0,gh}\mathbf{1})_-\|_2 \big\}, \tag{5}$$

with $u_+ = \max(0, u)$ and $u_- = \max(0, -u)$ entrywise. A regulator then enters for all basins of g or for none (group lasso, Yuan and Lin (2006)), with one sign across basins (cooperative lasso): an

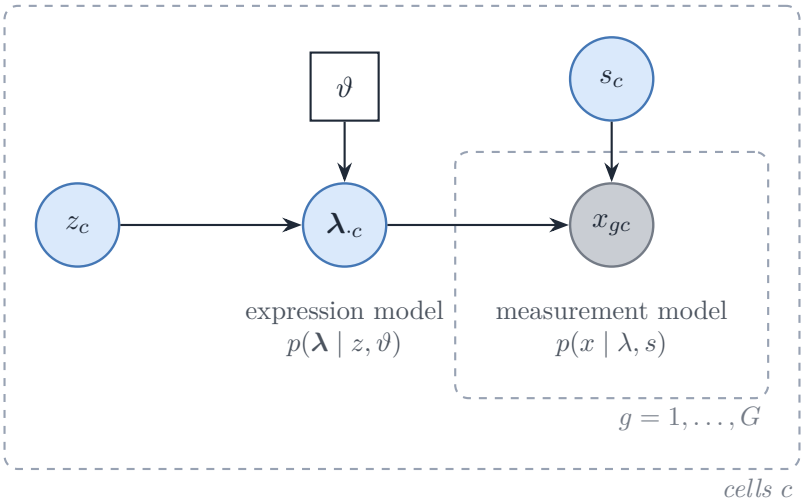


Figure 1: The two noise layers for one reference cell. z_c is the label (cell type j in DeCovarT, promoter basin in CardamomOT), ϑ the parameters ($\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_{W,j}$ in DeCovarT; NB shapes and rates in CardamomOT), s_c the depth factor. CardamomOT's paper models $\boldsymbol{\lambda} \mid z$ as a product of independent NB laws and omits s_c; its code adds s_c. DeCovarT estimates the covariance of x directly, so both layers enter $\boldsymbol{\Sigma}_{W,j}$.

activator stays an activator, and the B estimates of one edge share strength. The QUT carries over through the dual norm: with $\boldsymbol{s}_h$ the vector of the B null scores of edge $h \to g$, it is a quantile of $\max_h \max(\|\boldsymbol{s}_h^+\|_2, \|\boldsymbol{s}_h^-\|_2)/w_h$ for (5) (Chiquet, p. 80; our reading). Speculative: CardamomOT's $\boldsymbol{\theta}$ is shared across basins by construction (Eq. 1, p. 6), and basin-specific interactions need about B times more cells per edge.

In short: gene selection

- CardamomOT selects for change in time and multimodality, and treats the panel as prior information. DeCovarT selects for between-type separation with an estimable unimodal within-type law. The criteria are not interchangeable.
- The univariate ANOVA ratio is a screen; the Fisher information for $\boldsymbol{p}$ is the objective. For two types they agree, for every p, only with equal variances [V4c]; summing them over genes needs independent genes, except on a measure-zero set of correlations [V4e].
- DeCovarT's plug-in needs a well-defined covariance: the empirical one only from $C_j \geq G + 1$ with no exact relation between genes [V10a,V10b], shrinkage for any $C_j \geq 2$ [V11b]. A network reading needs the support of the precision, with $C_j - 1 \geq \mathrm{mlt}(\mathcal{G})$ for a refit and $C_j \gtrsim \deg^2_{\max} \log G$ for recovery. Prior structure (a whitelist, $\boldsymbol{\theta}_0$) or a joint penalty across classes borrows strength, the latter at the expense of rare types.
- CardamomOT's paper sets $\lambda = 0.05 N_c/\sqrt{G}$, a per-cell effect floor; its code uses a QUT at level 0.5 per target gene, a per-target FWER of 0.5 that, with about $\sqrt{G}$ regulators per gene, leaves a false-edge share near 6 % [V12c].

4 Expression model versus measurement model (axis ii)

Sarkar and Stephens (2021, p. 3) separate 'a measurement model [...] the conditional distribution $p(X \mid \Lambda)$' from 'a model for the expression levels $p(\Lambda)$'. For UMI data, Poisson measurement suffices: $x_{gc} \mid \lambda_{gc} \sim \mathrm{Poisson}(x_{c+}\lambda_{gc})$, and gene–gene correlation belongs to the expression model. Fang and Pachter (2025) add a cell-wide factor $c_{\text{tech}} c_{\text{bio}}$, our s_c, that multiplies the mean of every gene (their eqs 1–2, pp. 3–4). For intrinsically uncorrelated genes, the normalised covariance $\mathrm{Cov}(X_a, X_b)/(\mathbb{E}X_a \mathbb{E}X_b)$ then equals the extrinsic noise $\mathrm{Var}\, s/(\mathbb{E}s)^2$ (eq. 5, p. 5) [V9b]. They estimate it by the mode of the normalised covariances, call 'Poisson genes' those whose bootstrap interval of normalised variance contains it, and take the Poisson MLE of the size factor on those genes (p. 16). Normalising by size factors 'can eliminate biological stochasticity present in extrinsic "noise"' (p. 2).

CardamomOT, paper. The NB comes from bursting and is an expression model (p. 6). No size factor, library size or capture term appears in Eqs 3–4; technical noise is examined only as a robustness

test with simulated binomial dropout of 5–90 % (p. 17).

CardamomOT, code (version 2.0.0). The measurement layer is a per-cell depth factor s_c (`estimate_cell_depth.py`, `inference/depth.py`, `inference/depth_model.py`).

- **Rationale** (script docstring): a depth common to all genes of a cell 'acts like a hidden common regulator: it correlates all genes and can move cells between NB basins'.
- **Estimation** within homogeneous groups (sample, time, cell type), on the whole transcriptome (at least 10,000 genes, otherwise nothing is estimated); the library size leaves out the 50 most expressed genes. The default method, `poissonian`, follows Fang & Pachter: extrinsic noise as the mode of normalised covariances, iterated Poisson genes, s_c as the Poisson MLE on those genes, then a pairwise least-squares rescaling between groups. The alternative, `group_median`, divides the library size by the group median.
- **How it enters:** with the default `depth_model='mean'`, counts are $\mathrm{NB}(\kappa, r/s_c)$ (code: $\mathrm{NB}(k, c/s_i)$): depth multiplies the mean and leaves the shape κ unchanged, as in scVI. The option `'burst_frequency'` uses $\mathrm{NB}(\kappa s_c, r)$. Counts stay integers; s_c is stored in `obs['depth_factor']` and `X` is untouched.
- **When it is applied:** the diagnostic recommends the correction if the depth share of the mean within-group pairwise correlation exceeds 0.5, or if more than half of the genes have NB modes closer than the depth spread. The correlation share is computed on the log1p counts of 1,000 reference genes (the most variable among genes detected in more than 5 % of cells), with and without the log library size regressed out. The depth spread is the median over groups of $q_{0.95}/q_{0.05}$ of s_c. The correction is written only if the user allows it.

DeCovarT. $\boldsymbol{\Sigma}_{W,j}$ estimated from raw UMI counts mixes both layers. If $x_{gc} \mid \boldsymbol{\lambda}_{\cdot c}, s_c \sim \mathrm{Poisson}(s_c \lambda_{gc})$ with s_c independent of $\boldsymbol{\lambda}_{\cdot c}$ within type j, the law of total covariance gives (Appendix A)

$$\mathrm{Cov}(\boldsymbol{x}_{\cdot c}) = \mathbb{E}(s^2)\,\boldsymbol{\Sigma}_j^{\lambda} + \mathrm{Var}(s)\,\boldsymbol{\mu}_j^{\lambda}\boldsymbol{\mu}_j^{\lambda\mathsf{T}} + \mathbb{E}(s)\,\mathrm{diag}(\boldsymbol{\mu}_j^{\lambda}) \quad \text{[V9a]}. \tag{6}$$

- **Depth** adds a rank-one term along the mean $\boldsymbol{\mu}_j^{\lambda}$, common to all gene pairs. Its biological part (cell size, the S_j of DeCovarT's estimand) aggregates into the bulk and is wanted. Its technical part (capture) does not reach the bulk. Because the term lies along the type means, an inflated technical share down-weights the very directions that separate the types (our reading, not tested). Splitting the two parts needs controls such as spike-ins or ambient transcripts (Fang and Pachter 2025, pp. 8–11); CardamomOT's diagnostic measures the total depth share only.
- **Poisson sampling** adds to the diagonal only, so it attenuates every correlation and makes lowly expressed panels look independent.
- **Bulk scales** $c_i = a_i n_i$ and $\phi_i = a_i^2 n_i$ absorb a platform scale per bulk sample; they do not remove the cell-level depth variance from $\boldsymbol{\Sigma}_{W,j}$.
- **Raw or denoised in the bulk.** Let $\boldsymbol{\Sigma}_j^{\mathrm{bio}}$ be the covariance of the latent expression on the reference scale, so that the raw covariance is $\boldsymbol{\Sigma}_j^{\mathrm{bio}} + \mathrm{diag}(\boldsymbol{\mu}_j)$. If the bulk counts the pooled transcripts of its cells once, at depth a_i per reference unit, then $\mathrm{Cov}\,\boldsymbol{y}_{\cdot i} = \phi_i \sum_j p_{ji} \boldsymbol{\Sigma}_j^{\mathrm{bio}} + c_i\,\mathrm{diag}(\sum_j p_{ji}\boldsymbol{\mu}_j)$; binomial thinning of the reference counts gives the same law [V13a]. The reference Poisson diagonal is replaced by the bulk's own, of weight a_i^{-1} per unit of ϕ_i. The raw $\boldsymbol{\Sigma}_{W,j}$ is therefore exact only at $a_i = 1$, and the Poisson-denoised one only as $a_i \to \infty$. At any depth the plug-in is the denoised covariance plus $c_i\,\mathrm{diag}(\sum_j p_{ji}\boldsymbol{\mu}_j)$, which adds no parameter for UMI-like bulk counts [V13b,V13c]. The article reaches the same conclusion in its appendix on the within-type covariance that enters the bulk.
- **A model that separates the layers:** the PLN model, $x_{gc} \sim \mathrm{Poisson}(s_c e^{z_{gc}})$ with $\boldsymbol{z}_{\cdot c} \sim \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$ (Chiquet et al. 2021). Its offset is the measurement layer and $\boldsymbol{\Sigma}_z$ the expression layer. Mapped back to the linear scale, $\mathbb{E}\boldsymbol{x} = s\, e^{\boldsymbol{\mu}_z + \mathrm{diag}(\boldsymbol{\Sigma}_z)/2}$ and $\mathrm{Cov}\,\boldsymbol{x} = \mathrm{diag}(\mathbb{E}\boldsymbol{x}) + (\mathbb{E}\boldsymbol{x}\,\mathbb{E}\boldsymbol{x}^{\mathsf{T}}) \circ (e^{\boldsymbol{\Sigma}_z} - 1)$, with the exponential taken entrywise, for fixed s [V3a,V3b,V3c].

CardamomOT's s_c against ν^{tech}. CardamomOT's default depth estimator is Fang and Pachter's size factor on Poisson genes. It estimates the total factor $s_c = s_c^{\text{bio}} s_c^{\text{tech}}$, and the mode of normalised covariances estimates the total $\nu = \operatorname{Var} s$ (with $\mathbb{E}s = 1$), where $1 + \nu = (1 + \nu^{\text{bio}})(1 + \nu^{\text{tech}})$ [V14a]. DeCovarT's denoised covariance (vignette 05-1, `eq-denoised-within`) needs ν^{tech} alone. Given ν^{tech}, it is unbiased once $\bar{\boldsymbol{x}}\bar{\boldsymbol{x}}^{\mathsf{T}}$ is replaced by $\bar{\boldsymbol{x}}\bar{\boldsymbol{x}}^{\mathsf{T}} - \boldsymbol{V}_j/C_j$ [V14b,V14c,V14e]. Fed with the total ν, it returns $\boldsymbol{\Sigma}_j^{\lambda}$ and drops $\nu^{\text{bio}}(\boldsymbol{\Sigma}_j^{\lambda} + \boldsymbol{\mu}_j^{\lambda}\boldsymbol{\mu}_j^{\lambda\mathsf{T}})$, the cell-size share that a bulk sums [V14d]. CardamomOT's correction, which divides s_c out of the NB mean, is the count analogue of that total-ν plug-in. It suits a network of size-free expression; it does not suit a covariance that must reach a bulk (our reading).

Alternatives to the depth correction.

- **scVI** draws a library factor $\ell_n \sim$ LogNormal whose prior is the empirical mean and variance of the log library size in each batch, frequencies $\boldsymbol{\rho}_n$ that sum to one over genes, and counts Poisson$(\ell_n w_{ng})$ with $w_{ng} \sim \text{Gamma}(\rho_{gn}, \theta_g)$, plus zero inflation (Lopez et al. 2018, author manuscript, pp. 8–9). ℓ_n 'represents nuisance variation due to differences in capture efficiency and sequencing depth' (p. 3). It is the object s_c estimated inside a likelihood rather than on Poisson genes, and it also carries cell size because $\boldsymbol{\rho}_n$ sums to one. Genes are independent given a 10-dimensional latent (pp. 3, 8), so regulation would be read from that latent, not from a precision matrix (our reading).
- **scANVI** adds a cell-type label with a uniform prior, keeping the scVI library factor and count model (Xu et al. 2021, eqs 1–8, p. 15). For CardamomOT the label could be a type or a basin.
- **ComBat** corrects 'both additive and multiplicative batch effects' by empirical Bayes, per gene and batch (Johnson et al. 2007). Its count version ComBat-ref fits $\log \mu_{gij} = \alpha_g + \gamma_{gi} + \beta_{g,\text{cond}(j)} + \log N_j$ (gene g, batch i, sample j) and maps each count to a reference batch by matching negative binomial distribution functions (Zhang 2025, p. 2). It acts on samples, not cells. It could therefore replace CardamomOT's pairwise rescaling between groups, not s_c. With time points as batches it removes the time effect unless time enters the model as the biological condition.

In short: noise layers

- In the paper, CardamomOT's NB is an expression model with no measurement layer. The code of version 2.0.0 adds one: a per-cell depth s_c that multiplies the NB mean, applied only when depth explains more than half of the within-group correlation or blurs the NB basins.
- DeCovarT's $\boldsymbol{\Sigma}_{W,j}$ contains a rank-one depth term and a Poisson diagonal. The biological share of depth should stay, the technical share should not. CardamomOT's s_c is the total factor, so it cannot stand in for ν^{tech} [V14d].
- The raw $\boldsymbol{\Sigma}_{W,j}$ suits a bulk counted at the reference depth per cell ($a_i = 1$). Otherwise the bulk needs the denoised covariance plus its own Poisson term [V13b].
- DeCovarT can run the depth-share diagnostic on Suppinger before estimating $\boldsymbol{\Sigma}_{W,j}$.

5 The diagonal and self-regulation (axis iv)

Two different objects. In a Gaussian graphical model (GGM), $\omega_{gg} = 1/\operatorname{Var}(x_g \mid \boldsymbol{x}_{-g})$ is a conditional precision, and an edge is a non-zero partial correlation $-\omega_{gh}/\sqrt{\omega_{gg}\omega_{hh}}$. The diagonal is never an edge. In CardamomOT, θ_{gg} enters the activation rate of gene g through its own protein (Eq. 1, p. 6): a directed self-loop.

Self-activation and bistability. 'The combination of a positive auto-activation with a negative basal parameter β_i is the most parsimonious way to produce bistability for an isolated gene' (p. 28). This is the autocatalytic case of (Ferrell 2002): bistability needs feedback, 'some type of non-linearity within the feedback circuit', and balanced legs (p. 3 of that review). CardamomOT reports diagonal coefficients 'systematically non-negligible across all three datasets' (p. 18).

Linking the two through the stationary covariance. Around one basin, linearise the dynamics as an Ornstein–Uhlenbeck process $\mathrm{d}\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}^*)\mathrm{d}t + \boldsymbol{B}\,\mathrm{d}\boldsymbol{W}$. The stationary covariance solves the Lyapunov equation $\boldsymbol{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\boldsymbol{A}^{\mathsf{T}} + \boldsymbol{Q} = \boldsymbol{0}$, $\boldsymbol{Q} = \boldsymbol{B}\boldsymbol{B}^{\mathsf{T}}$. Write $a_g = \gamma_g - \varsigma_g > 0$ for the net decay of gene g: its degradation rate γ_g minus its linearised self-activation ς_g. For two genes with $1 \to 2$ of

strength b (Appendix A),

$$\sigma_{11} = \frac{q_1}{2a_1}, \qquad \sigma_{12} = \frac{b\,q_1}{2a_1(a_1+a_2)}, \qquad \omega_{12} = -\frac{2a_2(a_1+a_2)\,b}{(a_1+a_2)^2 q_2 + b^2 q_1} \quad \text{[V5a,V5b]}. \tag{7}$$

- Self-activation acts only through a_g: a larger ς_g inflates σ_{gg} and the conditional variance, so it lowers ω_{gg}. It appears nowhere as a 'self-edge'.
- $\omega_{12} \neq 0$ if and only if $b \neq 0$, but its sign and size mix b with the decays and noise intensities, and the direction is lost [V5c].
- With three genes, a fork $1 \to 2$, $1 \to 3$ gives $\omega_{23} \propto -b_{21}b_{31} \neq 0$: two co-regulated targets are linked in the precision although neither regulates the other [V5e]. A static structural equation model (SEM) with the same fork would give $\omega_{23} = 0$ [V5f].
- The linear model has one basin. Bistability requires the non-linearity that the linearisation removes, and produces a bimodal law within a cell type, which a Gaussian within-type law cannot represent. DeCovarT's counterpart of a basin is a subtype: more components in the convolution.

What the Lyapunov equation is for. A snapshot shows cells, not trajectories: it estimates the stationary law of the dynamics, never the dynamics themselves. Around one basin, $\boldsymbol{A}$ is the Jacobian of the regulation (off-diagonal: how gene h moves gene g; diagonal: minus the net decay a_g) and $\boldsymbol{Q}$ the intensity of the intrinsic noise (bursting, molecular noise). The Lyapunov equation is the map from these dynamic parameters to the second moment that a snapshot estimates, $(\boldsymbol{A}, \boldsymbol{Q}) \mapsto \boldsymbol{\Sigma}$. It is the bridge between the two estimands: CardamomOT's directed $\boldsymbol{\theta}$ lives in $\boldsymbol{A}$, DeCovarT's plug-in is $\boldsymbol{\Sigma}$. The map is not invertible: $\boldsymbol{\Sigma}$ has $G(G+1)/2$ free entries, $(\boldsymbol{A}, \boldsymbol{Q})$ up to $G^2 + G$, so many directed networks share one stationary covariance (our remark).

Lyapunov covariance against DeCovarT's denoised covariance. Add Poisson measurement at a fixed depth s, $x_{gc} \mid \boldsymbol{\lambda}_{\cdot c} \sim \text{Poisson}(s\lambda_{gc})$, with $\boldsymbol{\lambda}_{\cdot c}$ drawn from the stationary law of the basin. By Equation (6) with $\operatorname{Var} s = 0$,

$$\operatorname{Cov}(\boldsymbol{x}_{\cdot c}) = s^2 \boldsymbol{\Sigma}_{\text{Lyap}}(\boldsymbol{A}, \boldsymbol{Q}) + s \operatorname{diag}(\boldsymbol{\mu}^{\lambda}), \tag{8}$$

so DeCovarT's denoised covariance $\boldsymbol{\Sigma}_j^{\text{bio}}$ estimates $s^2\boldsymbol{\Sigma}_{\text{Lyap}}$, the stationary covariance, not $\boldsymbol{A}$. Both terms of Equation (8) have the form 'structured part plus diagonal', but the two diagonals act differently. The intrinsic noise $\boldsymbol{Q}$ enters the dynamics and is propagated by the network: a noise on gene 1 reaches gene 2 through b, which creates σ_{12} in Equation (7). The measurement diagonal is added after the dynamics and propagates nowhere. Reading $\boldsymbol{A}$ as the biological correlation and the measurement variance as $\boldsymbol{Q}$ would therefore conflate the two layers of Section 4 (our remark). One special case links the precision to the network directly: if $\boldsymbol{A}$ is symmetric and stable and $\boldsymbol{Q} = q\boldsymbol{I}$, then $\boldsymbol{\Sigma}_{\text{Lyap}} = -\frac{q}{2}\boldsymbol{A}^{-1}$ and $\boldsymbol{\Omega} = -\frac{2}{q}\boldsymbol{A}$ [V24a]. The off-diagonal of the precision is then the regulation up to sign and scale, and its diagonal is the net decay, $\omega_{gg} = 2a_g/q$. A symmetric $\boldsymbol{A}$ means reciprocal regulation of equal strength, a gradient system, which a gene regulatory network is in general not.

Why ω_{gg} is not a self-loop. For any positive-definite $\boldsymbol{\Omega}$, $\omega_{gg} = 1/\operatorname{Var}(x_g \mid \boldsymbol{x}_{-g}) = 1/\{\sigma_{gg}(1-R_g^2)\} > 0$, where R_g^2 is the squared multiple correlation of gene g on all the other genes [V24c]. The diagonal is always positive, so it cannot carry the sign of a self-loop (activation or repression). A large ω_{gg} says that gene g varies little once the other genes are known: either it is well predicted by them (R_g^2 close to one) or its own variance is small (strong decay, little noise). A small ω_{gg} says that gene g varies on its own. Self-activation enters only through that variance: it lowers a_g, raises σ_{gg} and lowers ω_{gg} [V5d]. CardamomOT's 'systematically non-negligible' self-loops (p. 18) and a large diagonal of $\hat{\boldsymbol{\Omega}}_j$ are therefore not the same finding.

A directed acyclic graph has a triangular factor. In a Gaussian Bayesian network, a linear SEM $\boldsymbol{x} = \boldsymbol{B}\boldsymbol{x} + \boldsymbol{e}$ with $\boldsymbol{e} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{D})$, $\boldsymbol{D}$ diagonal, and an acyclic graph, ordering the genes so that parents come first makes $\boldsymbol{B}$ strictly lower triangular, and

$$\boldsymbol{\Omega} = (\boldsymbol{I} - \boldsymbol{B})^{\mathsf{T}} \boldsymbol{D}^{-1} (\boldsymbol{I} - \boldsymbol{B}), \qquad \omega_{gg} = \frac{1}{d_g} + \sum_{h:\, g \to h} \frac{b_{hg}^2}{d_h}, \tag{9}$$

the modified Cholesky (LDL) factorisation of $\boldsymbol{\Omega}$, with a unit triangular factor $\boldsymbol{I}-\boldsymbol{B}$ [V24b] (Pourahmadi 1999). Acyclicity forces $b_{gg}=0$: a static directed acyclic graph has no self-loop, and its ω_{gg} is the noise precision of gene g plus what its children reveal about it. Every ordering of the genes gives such a factor, so the order, hence the direction of the edges, is not identified from $\boldsymbol{\Omega}$ alone (Markov equivalence); external information, such as CardamomOT's time points, has to supply it. A self-loop needs time: in a dynamic Bayesian network, $\boldsymbol{x}_t = \boldsymbol{A}\boldsymbol{x}_{t-1} + \boldsymbol{e}_t$, the entry A_{gg} is a lag-one effect of gene g on itself, and the graph unrolled over time stays acyclic. CardamomOT's snapshots are not time series of the same cells, so such a model would need its couplings between time points, as CardamomOT's transport supplies (our remark). Appendix D develops this statistical route: dynamic and mixed Gaussian Bayesian networks, structure learning and interventions.

Penalising the diagonal. When the graphical lasso penalises every entry, its covariance estimate has $\hat{W}_{gg} = V_{gg} + \lambda$ (Friedman et al. 2008): every variance is biased upwards by λ. DeCovarT's likelihood uses those variances as gene weights, so the diagonal is never penalised (`penalize.diagonal = FALSE`). For CardamomOT, the diagonal penalty is a prior choice. The paper says 'we deliberately choose not to apply a specific penalty on diagonal coefficients' (p. 28), and suggests zeroing entries of $\boldsymbol{\theta}_0$ when auto-regulation is implausible. Eq. 10 penalises $\boldsymbol{\theta}-\boldsymbol{\theta}_0$ entrywise. In the code (`network.py`), the first ℓ_1 step penalises the self-loop like any edge inside the prior support, and the adaptive-lasso refinement leaves 'self-regulation and basals unpenalized' (`refine_inference`). Stating that two-stage rule in the paper would remove an ambiguity.

In short: diagonal

- ω_{gg} is a conditional precision; θ_{gg} is a directed self-loop. They are related only through the stationary covariance of a linearised model, where self-activation lowers ω_{gg} [V5d].
- A precision edge is not a regulatory edge: a fork creates one without a direct link.
- The Lyapunov equation maps the dynamics $(\boldsymbol{A}, \boldsymbol{Q})$ to the stationary covariance a snapshot estimates; DeCovarT's denoised covariance estimates that covariance, not $\boldsymbol{A}$, and the measurement diagonal is added after the dynamics, whereas $\boldsymbol{Q}$ is propagated by them.
- $\omega_{gg} = 1/\{\sigma_{gg}(1-R_g^2)\}$ is always positive: a large diagonal means a gene well predicted by the others, not a self-loop [V24c]. A static SEM has a triangular factor and no self-loop [V24b]; self-regulation needs a dynamic model.
- Never penalise the GGM diagonal in DeCovarT. In CardamomOT the self-loop penalty is a prior choice, which the code makes differently in its two stages.

6 A side axis (iii): convolution, Wiener filter, mean field

Speculative

These are analogies to explore, not results. One item is a derivation: the Wiener filter is the best linear unbiased predictor (BLUP) of a mixed model.

- **Convolution and characteristic functions.** The bulk is a sum of independent cell profiles, so its density is a convolution, and its characteristic function is a product: $\varphi_{\boldsymbol{y}}(\boldsymbol{t}) = \prod_j \varphi_j^{\text{pop}}(p_j \boldsymbol{t})$ for population references. For single-cell references, with φ_j the characteristic function of one type-j cell, two models differ. Given the counts m_j of cells per type, $\varphi_{\boldsymbol{y}}(\boldsymbol{t}) = \prod_j \varphi_j(a_i \boldsymbol{t})^{m_j}$, and the covariance $\phi_i \sum_j p_j \boldsymbol{\Sigma}_{W,j}$, with $p_j = m_j/n_i$, is linear in $\boldsymbol{p}$: this is DeCovarT's single-cell model [V6b]. If each cell is instead drawn independently from the type mixture, $\varphi_{\boldsymbol{y}}(\boldsymbol{t}) = \varphi_{\text{mix}}(a_i \boldsymbol{t})^{n_i}$ with $\varphi_{\text{mix}} = \sum_j p_j \varphi_j$, and the covariance gains a between-type term that is quadratic in $\boldsymbol{p}$ [V6d,V6e] (Appendix A). DeCovarT's Gaussian is the central-limit form of that product. For a non-Gaussian cell law, such as a fitted CardamomOT NB mixture, the bulk law of one gene could be computed by fast Fourier transform (FFT) on the count lattice (Cooley and Tukey 1965). In several dimensions the lattice grows exponentially, so this stays a check for a few genes.
- **Wiener filter as BLUP.** Writing the type-j population profile of the sample as $\boldsymbol{\mu}_j + \boldsymbol{\varepsilon}_j$, with $\boldsymbol{\varepsilon}_j \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_j)$, makes $\boldsymbol{y} = \boldsymbol{\mu p} + \sum_j p_j \boldsymbol{\varepsilon}_j + \boldsymbol{e}$ a linear mixed model (LMM) whose random-effect design depends on $\boldsymbol{p}$. Its marginal covariance is $\sum_j p_j^2 \boldsymbol{\Sigma}_j + \boldsymbol{D}$ [V1a], and the BLUP of $\boldsymbol{\varepsilon}_j$ (Definitions box) is the Wiener filter of the DeCovarT article [V1b]. With $\boldsymbol{D} = \boldsymbol{0}$, maximum likelihood in $\boldsymbol{p}$, with the $\boldsymbol{\Sigma}_j$ plugged in, is DeCovarT's fit [V7a]; a nugget $\boldsymbol{D} \neq \boldsymbol{0}$, which DeCovarT leaves out (no extra measurement noise), changes the likelihood and moves $\hat{\boldsymbol{p}}$ [V7b] (Appendix A). Restricted maximum likelihood (REML) does not carry over directly: error contrasts remove $\boldsymbol{p}$ from the mean but not from $\boldsymbol{V}$, which depends on $\boldsymbol{p}$ [V7c]. No full Bayesian treatment is needed.
- **Mean field.** Hartree's ansatz writes a many-particle wave function as a product of one-particle functions, each in the average field of the others (Hartree 1928; Fock 1930). CardamomOT's product over genes given the basin is of that kind: genes interact only through the basin label and the protein field. DeCovarT's independent cells are another product ansatz. In both, the neglected correlation (Fock's exchange term in physics) is the within-basin covariance that CardamomOT reports losing (p. 12).

7 What each method could borrow

DeCovarT from CardamomOT.

- **The depth-share diagnostic**, run on Suppinger per (time, cell type) before $\boldsymbol{\Sigma}_{W,j}$ is estimated. A large share calls for a decision on the technical part of Equation (6), not for blind normalisation, since the biological part reaches the bulk.
- **Prior networks as penalty weights** (OmniPath, as in the CardamomOT repository), entering an edge-weighted graphical lasso.
- **A basin-separability check** within each type: a gene with separated NB modes inside a type signals a subtype that a Gaussian within-type law misrepresents.
- **The panel-as-prior stance** (p. 27): report the panel as part of the model, with the sensitivity of $\hat{\boldsymbol{p}}$ to it.

CardamomOT from DeCovarT.

- **A within-basin covariance.** The basin mixture implies a gene covariance that is diagonal plus a between-basin term of rank at most $B - 1$ for B basins, by the law of total covariance; the attenuation of the strongest correlations (p. 12) is the missing within-basin part. A Gaussian copula or a PLN layer within each basin would restore it.
- **A held-out likelihood for the penalty**, next to the quantile universal threshold. The expected Gaussian negative log-score $\frac{1}{2}\{\log\det \boldsymbol{\Sigma} + \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}_{\text{true}})\}$ is minimised at $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_{\text{true}}$ whatever the law of the cells, provided its second moments are finite and its mean is known [V2a,V2b,V2c,V2d], and cross-validation estimates the Kullback–Leibler risk for penalised estimators as well (Commenges and Jacqmin-Gadda 2015); (Commenges et al. 2012) give a fast approximation.
- **An information criterion for the panel**, aimed at the estimand $\boldsymbol{\theta}$ rather than at marginal

change in time.

- **Uncertainty for $\boldsymbol{\theta}$** beyond seed-to-seed variability, for instance a bootstrap over cells within time points.

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A Derivations

A.1 Two-gene Lyapunov equation. With $\boldsymbol{A} = \left(\begin{smallmatrix} -a_1 & 0 \\ b & -a_2 \end{smallmatrix}\right)$ and $\boldsymbol{Q} = \mathrm{diag}(q_1, q_2)$, the three distinct entries of $\boldsymbol{A\Sigma}+\boldsymbol{\Sigma A}^\mathsf{T}+\boldsymbol{Q} = \boldsymbol{0}$ read $-2a_1\sigma_{11}+q_1 = 0$, $b\sigma_{11}-(a_1+a_2)\sigma_{12} = 0$ and $2b\sigma_{12}-2a_2\sigma_{22}+q_2 = 0$. Hence σ_{11} and σ_{12} as in Equation (7) and $\sigma_{22} = \{a_1(a_1+a_2)q_2 + b^2q_1\}/\{2a_1a_2(a_1+a_2)\}$. Inverting gives

$$\omega_{11} = \frac{2\{a_1(a_1+a_2)^2q_2 + (a_1+a_2)b^2q_1\}}{q_1\{(a_1+a_2)^2q_2 + b^2q_1\}}, \qquad \omega_{22} = \frac{2a_2(a_1+a_2)^2}{(a_1+a_2)^2q_2 + b^2q_1},$$

and ω_{12} of Equation (7). For the fork $1 \to 2$ (strength b), $1 \to 3$ (strength c), $\omega_{23} = -4a_2a_3bc\,q_1(a_1 + a_2)(a_1 + a_3)(a_2 + a_3)/\mathcal{D}$ with $\mathcal{D} > 0$ a polynomial in the rates and noise intensities [V5e]: $\mathcal{D}$ is $b^2c^2q_1^2(a_2 - a_3)^2$ plus a polynomial with positive coefficients. The ledger derives all these forms from the Lyapunov equation [V5a,V5b].

A.2 Gaussian convolution. Population references: with independent type-j population profiles $\boldsymbol{x}_j^{\mathrm{pop}} \sim \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$, the bulk $\boldsymbol{y} = \sum_j p_j\boldsymbol{x}_j^{\mathrm{pop}}$ has $\varphi_{\boldsymbol{y}}(\boldsymbol{t}) = \prod_j \varphi_j^{\mathrm{pop}}(p_j\boldsymbol{t}) = \exp\{\mathrm{i}\,\boldsymbol{t}^\mathsf{T}\sum_j p_j\boldsymbol{\mu}_j - \frac{1}{2}\boldsymbol{t}^\mathsf{T}(\sum_j p_j^2\boldsymbol{\Sigma}_j)\boldsymbol{t}\}$, hence the quadratic covariance. Single-cell references (one type-j cell: mean $\boldsymbol{\mu}_j$, covariance $\boldsymbol{\Sigma}_{W,j}$, characteristic function φ_j) admit two models. (a) Given m_j cells of type j, the bulk has characteristic function $\prod_j \varphi_j(a_i\boldsymbol{t})^{m_j}$, mean $a_i\sum_j m_j\boldsymbol{\mu}_j = c_i\sum_j p_j\boldsymbol{\mu}_j$ and covariance $a_i^2\sum_j m_j\boldsymbol{\Sigma}_{W,j} = \phi_i\sum_j p_j\boldsymbol{\Sigma}_{W,j}$, with the realised fractions $p_j = m_j/n_i$; it is linear in $\boldsymbol{p}$ [V6b]. This conditional model is DeCovarT's. (b) If each of the n_i cells is drawn independently from the type mixture, one cell has characteristic function $\varphi_{\mathrm{mix}} = \sum_j p_j\varphi_j$ and the bulk has $\varphi_{\mathrm{mix}}(a_i\boldsymbol{t})^{n_i}$, the average of the product in (a) over $\boldsymbol{m} \sim \mathrm{Multinomial}(n_i, \boldsymbol{p})$. The law of total covariance then adds $a_i^2n_i\boldsymbol{\mu}\{\mathrm{diag}(\boldsymbol{p}) - \boldsymbol{pp}^\mathsf{T}\}\boldsymbol{\mu}^\mathsf{T}$ to the covariance of (a); at $J = 2$ this is $\phi_ip(1-p)\boldsymbol{\Delta\Delta}^\mathsf{T}$, quadratic in p [V6d,V6e].

A.3 Wiener filter as BLUP. Stack $\boldsymbol{u} = (\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_J)$, with covariance $\boldsymbol{\Gamma} = \mathrm{blockdiag}(\boldsymbol{\Sigma}_j)$, and let $\boldsymbol{Z} = (p_1\boldsymbol{I}_G, \ldots, p_J\boldsymbol{I}_G)$. Then $\boldsymbol{y} = \boldsymbol{\mu p} + \boldsymbol{Zu} + \boldsymbol{e}$, $\boldsymbol{e} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{D})$, and $\boldsymbol{V} = \mathrm{Var}\,\boldsymbol{y} = \boldsymbol{Z\Gamma Z}^\mathsf{T} + \boldsymbol{D} = \sum_j p_j^2\boldsymbol{\Sigma}_j + \boldsymbol{D}$. The BLUP is $\hat{\boldsymbol{u}} = \boldsymbol{\Gamma Z}^\mathsf{T}\boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{\mu p})$, whose block j is $p_j\boldsymbol{\Sigma}_j\boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{\mu p})$: the Wiener filter of the DeCovarT article (§ Wiener filter), equal to the posterior mean in the Gaussian model [V1b,V1c]. Integrating $\boldsymbol{u}$ out gives $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu p}, \boldsymbol{V})$. At $\boldsymbol{D} = \boldsymbol{0}$ its log-likelihood in $\boldsymbol{p}$ is DeCovarT's Gaussian log-likelihood [V7a]; with $\boldsymbol{D} \neq \boldsymbol{0}$ the maximiser moves, from 0.747 to 0.749 in a one-gene example with unit nugget [V7b]. For $J = 2$, the error contrasts $\boldsymbol{K}^\mathsf{T}\boldsymbol{y}$ with $\boldsymbol{K}^\mathsf{T}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) = \boldsymbol{0}$ have covariance $\boldsymbol{K}^\mathsf{T}\boldsymbol{V}(\boldsymbol{p})\boldsymbol{K}$, which still depends on $\boldsymbol{p}$, so REML does not separate the fixed effect from the variance here [V7c].

A.4 ANOVA ratio and Fisher information, $J = 2$. One gene, single-cell model: $y \sim \mathcal{N}(c\{p\mu_1 + (1-p)\mu_2\}, \phi\{p\sigma_1^2 + (1-p)\sigma_2^2\})$, with $c = c_i$ and $\phi = \phi_i$. For $\mathcal{N}(m(p), v(p))$, $I(p) = m'(p)^2/v + \frac{1}{2}\{v'(p)/v\}^2$, so $I(p) = c^2\Delta^2/v + \frac{1}{2}\{(\sigma_1^2 - \sigma_2^2)/(p\sigma_1^2 + (1-p)\sigma_2^2)\}^2$. If $\sigma_1^2 = \sigma_2^2 = \sigma^2$, the second term vanishes and $I(p) = (c^2/\phi)\Delta^2/\sigma^2 = n_i\Delta^2/\sigma^2$, since $c^2/\phi = n_i$. With $C_1 = C_2 = C$ reference cells per type, the ANOVA ratio is the plug-in estimate of $(C/2)\Delta^2/\sigma^2$, but its mean is $\{\nu/(\nu-2)\}\{1 + (C/2)\Delta^2/\sigma^2\}$ with $\nu = 2C - 2$, so $I(p) \approx (2n_i/C)(F-1)$ for large C [V4b]. Over independent genes the Fisher information terms add; with correlated genes they do not in general: for two genes with correlation ρ they add only at $\rho = 0$ or $\rho = 2d_1d_2/(d_1^2 + d_2^2)$, with $d_g = \Delta_g/\sigma_g$ [V4e]. Under the population model, $v(p) = p^2\sigma_1^2 + (1-p)^2\sigma_2^2$ depends on p even when $\sigma_1 = \sigma_2$, and the proportionality fails: $I(p)$ becomes affine in Δ^2/σ^2, with an offset common to all genes [V4f].

Why one gene first, and what changes with several. The ANOVA screen ranks genes one at a time, so the comparison above is made gene by gene. DeCovarT's objective is multivariate. For a panel of G genes with type covariances $\boldsymbol{\Sigma}_{W,1}$ and $\boldsymbol{\Sigma}_{W,2}$, the single-cell model reads $\boldsymbol{y} \sim \mathcal{N}\{c\,\boldsymbol{m}(p), \phi\boldsymbol{S}(p)\}$ with $\boldsymbol{S}(p) = p\boldsymbol{\Sigma}_{W,1} + (1-p)\boldsymbol{\Sigma}_{W,2}$, and the Gaussian Fisher information $\dot{\boldsymbol{m}}^\mathsf{T}\boldsymbol{V}^{-1}\dot{\boldsymbol{m}} + \frac{1}{2}\,\mathrm{tr}\{(\boldsymbol{V}^{-1}\dot{\boldsymbol{V}})^2\}$ (dots for derivatives in p) gives

$$I(p) = n_i\,\boldsymbol{\Delta}^\mathsf{T}\boldsymbol{S}(p)^{-1}\boldsymbol{\Delta} + \tfrac{1}{2}\,\mathrm{tr}\Big[\big\{\boldsymbol{S}(p)^{-1}(\boldsymbol{\Sigma}_{W,1} - \boldsymbol{\Sigma}_{W,2})\big\}^2\Big] \quad \text{[V23a]}. \tag{10}$$

The first term is the squared Mahalanobis distance between the type means in the metric of the mixed covariance; with a common $\boldsymbol{\Sigma}_W$ it is V4d. Correlations change it in either direction: two genes with the same Δ_g/σ_g and a positive within-type correlation carry less than twice the information of one (V4e). The second term rewards covariances that differ between the types, whatever the means. It does not grow with n_i, so in a bulk of many cells the covariance acts mainly through the metric $\boldsymbol{S}(p)^{-1}$, as a GLS weight, and only marginally as a separate source of information (our remark).

The same two ingredients define the overlap of the two type laws, the criterion of the screening scenario. The Bhattacharyya coefficient $\int \sqrt{f_1 f_2} = e^{-D_B} = 1 - H^2$, with H the Hellinger distance, has

$$D_B = \tfrac{1}{8}\boldsymbol{\Delta}^{\mathsf{T}}\bar{\boldsymbol{\Sigma}}^{-1}\boldsymbol{\Delta} + \tfrac{1}{2}\log\frac{\det\bar{\boldsymbol{\Sigma}}}{\sqrt{\det\boldsymbol{\Sigma}_{W,1}\det\boldsymbol{\Sigma}_{W,2}}}, \qquad \bar{\boldsymbol{\Sigma}} = \tfrac{1}{2}(\boldsymbol{\Sigma}_{W,1} + \boldsymbol{\Sigma}_{W,2}) \quad \text{[V23b]}, \tag{11}$$

which reduces to V8a when the covariances are equal. Minimising the overlap therefore maximises a mean separation in a covariance metric plus a covariance-difference term, the multivariate counterpart of the one-gene ratio. The two criteria differ in their object: the overlap is a property of the type laws in the reference, Equation (10) of the composition of a bulk of n_i cells.

A.5 Depth and sampling in the within-type covariance. Conditionally on $(\boldsymbol{\lambda}, s)$, counts are independent Poisson with means $s\boldsymbol{\lambda}$. Then $\mathbb{E}\boldsymbol{x} = \mathbb{E}(s)\boldsymbol{\mu}^{\lambda}$ and $\operatorname{Cov}\boldsymbol{x} = \mathbb{E}\{\operatorname{Cov}(\boldsymbol{x} \mid \boldsymbol{\lambda}, s)\} + \operatorname{Cov}\{\mathbb{E}(\boldsymbol{x} \mid \boldsymbol{\lambda}, s)\} = \mathbb{E}(s)\operatorname{diag}(\boldsymbol{\mu}^{\lambda}) + \operatorname{Cov}(s\boldsymbol{\lambda})$, with $\operatorname{Cov}(s\boldsymbol{\lambda}) = \mathbb{E}(s^2)\boldsymbol{\Sigma}^{\lambda} + \operatorname{Var}(s)\boldsymbol{\mu}^{\lambda}\boldsymbol{\mu}^{\lambda\mathsf{T}}$ when s and $\boldsymbol{\lambda}$ are independent; this is Equation (6) [V9a]. Dividing an off-diagonal term by the product of the two means gives Fang & Pachter's normalised covariance $\operatorname{Var}(s)/(\mathbb{E}s)^2$ for genes with $\Sigma^{\lambda}_{gh} = 0$, whatever the scale of s; the same ratio is the normalised variance $\{\operatorname{Var}(x_g) - \mathbb{E}x_g\}/(\mathbb{E}x_g)^2$ of a gene with $\Sigma^{\lambda}_{gg} = 0$, one of their Poisson genes (their eq. 5) [V9b].

B Verification ledger

Each statement checked by SymPy carries its ledger identifier. The scripts are in `verification/sympy/`; `verification/run_all.py` runs them and writes the table below. V1: LMM covariance and BLUP = Wiener filter; V2: Stein's loss minimised at the true covariance; V3: Poisson log-normal moment map; V4: Fisher information against the ANOVA ratio for $J = 2$; V5: two-gene Lyapunov covariance and precision, fork precision; V6: single-cell linear against population quadratic covariance at $J = 2$, and the two single-cell models (given counts, or i.i.d. draws); V7: maximum likelihood of the mixed model against DeCovarT's fit, and REML; V8: Hellinger and 2-Wasserstein distances against the Mahalanobis distance; V9: depth and Poisson sampling in the within-type covariance (Equation (6)) and the normalised covariance. Each check derives the result (Gaussian conditioning and integration, cumulant generating functions, score and Fisher information, the Lyapunov equation, couplings, the law of total covariance) and compares it with the printed formula. Probabilistic statements hold under the stated assumptions: Gaussian laws (V1, V4, V6, V7, V8), finite second moments and a known mean (V2), a fixed offset s (V3), linearised dynamics around one basin (V5), given cell counts (V6b) or i.i.d. draws from the type mixture (V6d, V6e), and Poisson counts with s independent of $\boldsymbol{\lambda}$ and finite second moments (V9). Six entries are flags, each corrected in the text: the entrywise exponential (V3b), the mean of the ANOVA ratio (V4b), the exceptional correlations at which per-gene information adds up (V4e), the nugget $\boldsymbol{D}$ under which maximum likelihood is no longer DeCovarT's fit (V7b), REML (V7c), and the needless normalisation $\mathbb{E}s = 1$ (V9b). The ledger writes n for the number n_i of cells in a bulk sample, a for a_i, and C for the common number of reference cells per type in V4.

V10: rank of the empirical covariance $\boldsymbol{V}_j$ (symbolic and rational data), rank loss under exact relations, and the unbounded likelihood when $\boldsymbol{V}_j$ is singular; V11: positive definiteness of linear shrinkage (2×2 polynomial identity, quadratic-form argument for any G, a numeric check at $G = 40$, $C = 2$); V12: the quantile universal threshold against the Bonferroni and Šidák thresholds, the per-target FWER, the expected false edges and the false-edge share, and the scaling of CardamomOT's λ. V12 assumes independent Gaussian null scores and true scores far above the threshold, and writes m for R and k for R_1; V10 writes C and G for C_j and the panel size. One more entry is a flag: the off-by-one sample-size condition of Zwiernik et al. (V10d).

V13: bulk covariance under three sampling models (cell-depth counts scaled by a, bulk-depth Poisson sampling, binomial thinning), from a generic third-order cumulant generating function per type, and the toy example of vignette 02-1; V14: the denoised within-type covariance of vignette 05-1 under Poisson counts with independent size factors of mean one, $s^{\text{bio}} \perp s^{\text{tech}} \perp \boldsymbol{\lambda}$, and i.i.d. cells, with a Monte Carlo confirmation; V15: the Gaussian transport map of vignette 05-1 (Brenier's optimality theorem is quoted, not checked). One more flag: the finite-sample bias of the denoised estimator as first printed (V14c), corrected in vignette 05-1. Vignettes cite identifiers in HTML comments; the table writes `qmd` for an identifier cited only there, and `ledger.json` lists the vignette lines (key `qmd_lines`).

V16: identifiability of the bulk sampling diagonal against the variance scale, cited in vignette 01-2. V17–V19 are cited in the feature-selection vignette 03-1. V17: what the presence gate excludes, from symbolic counts: a gene with no count in one type gives a zero row of $\boldsymbol{V}_j$, a zero determinant and a 0/0 correlation; a gene detected in one cell only has a correlation set by that cell, equal to one for two such genes. V18: the D-, E- and A-criteria of gene-panel selection under the single-cell model with a common within-type covariance, where the information is $n_i \boldsymbol{K}^{\mathsf{T}} \boldsymbol{M}^{\mathsf{T}} \boldsymbol{\Sigma}_W^{-1} \boldsymbol{M} \boldsymbol{K}$ (for $J = 2$ this is V4d); the Gram volume and $\kappa(\boldsymbol{I}) = \kappa(\boldsymbol{M}\boldsymbol{K})^2$ when $\boldsymbol{\Sigma}_W = \boldsymbol{I}$; the dependence of E and A on the simplex basis; and a two-panel example in which a within-type correlation makes D and E disagree. V19: an explicit tail bound for RNA-Sieve's range filter. RNA-Sieve states only that the tail is sub-Gaussian; the bound, from Hoeffding's inequality (quoted, not checked) and a union bound for expression in $[0, B]$, is ours, with a Monte Carlo confirmation.

V20–V22 are cited in vignettes 03-1 and S5: sub-exponential negative-binomial counts and Chernoff and Bernstein versions of the range bound (V20), Unico's confounded variance components and the spectral entropy of a correlation matrix (V21), and the moments of CSNet's shared-gamma negative-

binomial simulator with the Woodbury form of a factor precision (V22). V23: the multivariate Fisher information for p at $J = 2$, derived from the Gaussian log-likelihood, and the Bhattacharyya distance between two Gaussian type laws, checked against numerical integration. V24: the symmetric case of the Lyapunov equation, the modified Cholesky factor of the precision of a linear SEM on a directed acyclic graph, and the conditional precision as $1/\{\sigma_{gg}(1 - R_g^2)\}$. V25 is cited in the French summary of this note (intrinsic noise, bulk scales, naive intervals). V26: the discrete Lyapunov equation of a linear Gaussian DBN with a feedback loop, the rotation that leaves the moments of two snapshots unchanged, the covariance of a mixture of classes, two I-equivalent networks that differ under intervention, and the precision of a v-structure (Appendix D).

Generated 2026-11-02 with SymPy 1.14.0, NumPy 2.5.4, Python 3.14.3. 91 checks: 83 pass, 8 flag, 0 fail. 'flag' means the engine confirms a deviation from the original wording, now corrected in the text. The text line is where the identifier is cited; `ledger.json` keeps the full engine output of each check (key `detail_full`).

ID	Claim	Detail	Kind	Status	Line
V1	*Linear mixed model: covariance, BLUP and Wiener filter*				
V1a	Marginal covariance of the mixed model: $\operatorname{Var}\boldsymbol{y} = \sum_j p_j^2 \boldsymbol{\Sigma}_j + \boldsymbol{D}$	$\boldsymbol{y} - \boldsymbol{\mu p} = \boldsymbol{M w}$ with $\boldsymbol{w}$ standard normal; $\boldsymbol{M M}^{\mathsf{T}}$ equals the printed form entrywise ($G = J = 2$, symbolic Cholesky factors)	exact	pass	959
V1b	BLUP (= MAP of $\boldsymbol{\varepsilon}_j$ given $\boldsymbol{p}$) is $p_j \boldsymbol{\Sigma}_j \boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{\mu p})$, the Wiener filter	the printed predictor solves Henderson's mixed-model equations identically; their matrix is positive definite, so it is the unique BLUP and MAP	exact	pass	960, 1054
V1c	Posterior mean $\mathbb{E}(\boldsymbol{\varepsilon}_j \mid \boldsymbol{y}, \boldsymbol{p})$ equals the BLUP in the Gaussian model	$\operatorname{Cov}(\boldsymbol{u}, \boldsymbol{y})$ from the linear map has blocks $p_j \boldsymbol{\Sigma}_j$; Gaussian conditioning returns the predictor of V1b	exact	pass	1054
V2	*Stein's loss minimised at the true covariance*				
V2a	Scalar case: $L(s) = \frac{1}{2}(\log s + t/s)$ has its unique minimum at $s = t$	$L'(s) = 0$ only at $s = t$; $L''(t) = \frac{1}{2t^2} > 0$; $L(s) - L(t) = \frac{1}{2}(x - 1 - \log x) \geq 0$ with $x = t/s$	exact	pass	1000
V2b	2×2: the gradient of L vanishes at $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_{\text{true}}$ and the Hessian is positive definite there	gradient zero at $\boldsymbol{\Sigma}_{\text{true}} = \boldsymbol{L L}^{\mathsf{T}}$ (symbolic Cholesky factor); leading principal minors of the Hessian $\frac{\left(l_{21}^2 + l_{22}^2\right)^2}{2 l_{11}^4 l_{22}^4}$, $\frac{\left(l_{21}^2 + l_{22}^2\right)^2}{2 l_{11}^6 l_{22}^6}$, $\frac{1}{4 l_{11}^6 l_{22}^6}$, all positive	exact	pass	1000
V2c	Numeric: $L(\boldsymbol{\Sigma}) > L(\boldsymbol{\Sigma}_{\text{true}})$ at random SPD $\boldsymbol{\Sigma}$	mpmath at 40 digits, seed 20261030, $d = 4$, 20 true matrices with 50 random competitors each; smallest gap 0.93808	numeric	pass	1000
V2d	The expected Gaussian log-score depends on the cell law only through its covariance, so the minimiser is $\boldsymbol{\Sigma}_{\text{true}}$ for any law with finite second moments	centred 4-point non-Gaussian law: expected score $= \log 2\pi + L(\boldsymbol{\Sigma})$; gradient zero at its covariance $\begin{bmatrix} \frac{49}{18} & \frac{31}{36} \\ \frac{31}{36} & \frac{395}{144} \end{bmatrix}$; the mean is taken as known	exact	pass	1000
V3	*Poisson log-normal moment map*				
V3a	Poisson log-normal moments for fixed s: $\mathbb{E}\boldsymbol{x} = s\, e^{\boldsymbol{\mu} + \operatorname{diag}(\boldsymbol{\Sigma})/2}$, $\operatorname{Cov}\boldsymbol{x} = \operatorname{diag}(\mathbb{E}\boldsymbol{x}) + (\mathbb{E}\boldsymbol{x}\,\mathbb{E}\boldsymbol{x}^{\mathsf{T}}) \circ (e^{\boldsymbol{\Sigma}} - 1)$, exponential entrywise	Poisson moments summed from the pmf ($\mathbb{E}X = \lambda$, $\mathbb{E}X(X-1) = \lambda^2$); normal MGF $e^{t\left(m + \frac{tv}{2}\right)}$ integrated from the density; tower rule, $G = 2$	exact	pass	761
V3b	$e^{\boldsymbol{\Sigma}_z}$ in the covariance is the entrywise exponential, not the matrix exponential	for $\boldsymbol{\Sigma} = \left(\begin{smallmatrix} 0.5 & 0.2 \\ 0.2 & 0.3 \end{smallmatrix}\right)$, entrywise minus matrix exponential $= \begin{bmatrix} -0.031 & 0.921 \\ 0.921 & -0.029 \end{bmatrix}$; the derivation uses $e^{\Sigma_{gh}}$ entry by entry	numeric	**flag**	761
V3c	Monte Carlo confirmation of the Poisson log-normal moments ($G = 3$, fixed s)	NumPy seed 20261030, $n = 2{,}000{,}000$, $s = 0.8$; largest $\lvert z \rvert$ over 3 means and 6 covariances $= 0.99$ (tolerance 4.5)	numeric	pass	761
V4	*Fisher information against the ANOVA ratio, $J = 2$*				
V4a	Single-cell model, $J = 2$, one gene: $I(p) = c^2\Delta^2/v + \frac{1}{2}\{(\sigma_1^2 - \sigma_2^2)/v\}^2$, and $I(p) = n_i\Delta^2/\sigma^2$ when $\sigma_1 = \sigma_2$	$-\mathbb{E}\,\partial_p^2 \ell$ with $y = m + \sqrt{v}\,z$ equals the printed form; with equal variances, $c = an$ and $\phi = a^2 n$: $I = \frac{n(\mu_1 - \mu_2)^2}{\sigma^2}$	exact	pass	265
V4b	ANOVA ratio, C cells per type: F is the plug-in of $(C/2)\Delta^2/\sigma^2$, but $\mathbb{E}F = \frac{\nu}{\nu - 2}\{1 + (C/2)\Delta^2/\sigma^2\}$	$\mathrm{MS}_B = C(\bar{y}_1 - \bar{y}_2)^2/2$ on symbolic data; $\mathbb{E}(1/\chi_4^2) = \frac{1}{2}$; at $C = 3$, $\mathbb{E}F = \frac{3\mu_1^2 - 6\mu_1\mu_2 + 3\mu_2^2 + 2\sigma^2}{\sigma^2}$. Deviation: $I(p) \approx (2n_i/C)(F - 1)$, not $(2n_i/C)F$	exact	**flag**	267, 1069
V4c	Fisher information and ANOVA ratio agree for every p ($J = 2$) only with equal within-type variances	$I(p) - n\Delta^2/\bar{s}^2$ is a polynomial in p whose coefficients all vanish only at $S_2 = S_1$ ($\Delta \neq 0$)	exact	pass	645
V4d	Multivariate information with a common $\boldsymbol{\Sigma}_W$ is $n_i \boldsymbol{\Delta}^{\mathsf{T}} \boldsymbol{\Sigma}_W^{-1} \boldsymbol{\Delta}$	bivariate $-\mathbb{E}\,\partial_p^2 \ell$, expectation over $\boldsymbol{y} = \boldsymbol{m} + \sqrt{\phi}\,\boldsymbol{L z}$, equals $n \boldsymbol{\Delta}^{\mathsf{T}} \boldsymbol{\Sigma}_W^{-1} \boldsymbol{\Delta}$	exact	pass	269

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ID	Claim	Detail	Kind	Status	Line
V4e	Per-gene informations add for independent genes; with correlated genes, only at isolated values of ρ	n^{-1}(multivariate − sum of per-gene) $= -\frac{\rho\left(d_1^2\rho - 2d_1d_2 + d_2^2\rho\right)}{(\rho-1)(\rho+1)}$, zero at $\rho \in \{0, \frac{2d_1d_2}{d_1^2+d_2^2}\}$. Deviation: additivity also holds at the second root, a measure-zero set	exact	**flag**	646, 1071
V4f	Population model, equal variances: $v(p)$ depends on p, and $I(p)$ is not proportional to the ANOVA ratio	$I(p) = \frac{\Delta^2}{\sigma^2} \cdot \frac{1}{2p^2-2p+1} + \frac{2(2p-1)^2}{(2p^2-2p+1)^2}$; the offset is free of Δ and σ, so the ranking at fixed p is kept	exact	pass	1073
V5	*Two-gene Lyapunov covariance and precision; fork precision*				
V5a	Two-gene Lyapunov solution: σ_{11}, σ_{12} (eq:lyap2) and σ_{22} (A.1)	$\boldsymbol{AS} + \boldsymbol{SA}^\mathsf{T} + \boldsymbol{Q} = \boldsymbol{0}$ solved as a linear system: $\sigma_{12} = \frac{bq_1}{2a_1(a_1+a_2)}$, $\sigma_{22} = \frac{a_1^2q_2 + a_1a_2q_2 + b^2q_1}{2a_1a_2(a_1+a_2)}$; equal to the printed forms	exact	pass	837, 1030
V5b	Two-gene precision: ω_{12} (eq:lyap2), ω_{11} and ω_{22} (A.1)	$\boldsymbol{S}^{-1}$ from the engine's $\boldsymbol{S}$: $\omega_{12} = -\frac{2a_2b(a_1+a_2)}{a_1^2q_2 + 2a_1a_2q_2 + a_2^2q_2 + b^2q_1}$; all three equal the printed forms	exact	pass	837, 1030
V5c	$\omega_{12} \neq 0$ if and only if $b \neq 0$	numerator of ω_{12}: $-2a_2b\,(a_1 + a_2)$, only root in b: 0; denominator $(a_1 + a_2)^2q_2 + b^2q_1 > 0$	exact	pass	843
V5d	ω_{gg} falls as self-activation rises (a_g falls), and depends on the edge and on the other gene	$\partial\omega_{11}/\partial a_1$ and $\partial\omega_{22}/\partial a_2$ have numerators and denominators with positive coefficients; $\partial\omega_{22}/\partial a_1 = \frac{4a_2b^2q_1(a_1+a_2)}{(a_1^2q_2 + 2a_1a_2q_2 + a_2^2q_2 + b^2q_1)^2}$	exact	pass	886, 922
V5e	Fork $1 \to 2$ (b), $1 \to 3$ (c): $\omega_{23} = -4a_2a_3bc\,q_1(a_1 + a_2)(a_1 + a_3)(a_2 + a_3)/\mathcal{D}$ with $\mathcal{D} > 0$	six-unknown Lyapunov system solved exactly; $\mathcal{D} = b^2c^2q_1^2(a_2 - a_3)^2$ plus a polynomial with positive coefficients, so $\mathcal{D} > 0$	exact	pass	846, 1029
V5f	Static structural equation model with the same fork: $\omega_{23} = 0$	$(\boldsymbol{I} - \boldsymbol{B})^{-1}\boldsymbol{D}(\boldsymbol{I} - \boldsymbol{B})^{-\mathsf{T}}$ inverted exactly: $\omega_{23} = 0$, $\omega_{12} = -\frac{b}{d_2}$	exact	pass	846
V6	*Single-cell (linear) against population (quadratic) covariance*				
V6a	Population references: $\operatorname{Cov}\boldsymbol{y} = \sum_j p_j^2\boldsymbol{\Sigma}_j$, quadratic in $\boldsymbol{p}$	Hessian at $\boldsymbol{0}$ of $K_{\boldsymbol{y}}(\boldsymbol{t}) = \sum_j K_j(p_j\boldsymbol{t})$ is $p^2\boldsymbol{\Sigma}_1 + (1-p)^2\boldsymbol{\Sigma}_2$; $\partial_p^2 = 2(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2) \succ 0$	exact	pass	178
V6b	Single-cell references, given $m_j = n_ip_j$ cells per type: $\operatorname{Cov}\boldsymbol{y} = \phi_i\sum_j p_j\boldsymbol{\Sigma}_{W,j}$, linear in $\boldsymbol{p}$, $\phi_i = a_i^2n_i$	Hessian at $\boldsymbol{0}$ of $\sum_j m_jK_j(a\boldsymbol{t})$ is $a^2n\{p\boldsymbol{\Sigma}_1 + (1-p)\boldsymbol{\Sigma}_2\}$; $\partial_p^2 = \boldsymbol{0}$; conditional on the counts m_j	exact	pass	948, 1039
V6c	At $J = 2$ the quadratic and the linear covariance differ for any common scale k	the p^2 coefficient of $\sum_j p_j^2\boldsymbol{\Sigma}_j - k\sum_j p_j\boldsymbol{\Sigma}_j$ is $\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2$, with a positive diagonal; equality only at isolated p	exact	pass	178
V6d	Cells drawn i.i.d. from the type mixture: $\operatorname{Cov}\boldsymbol{y}$ gains $a^2np(1-p)\boldsymbol{\Delta}\boldsymbol{\Delta}^\mathsf{T}$, quadratic in p	Hessian at $\boldsymbol{0}$ of $n\log\sum_j p_je^{K_j(a\boldsymbol{t})}$ minus the fixed-count covariance equals $a^2np(1-p)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\mathsf{T}$	exact	pass	951, 1045
V6e	With i.i.d. draws, the bulk characteristic function is $\varphi_{\text{mix}}(\boldsymbol{t})^n$, $\varphi_{\text{mix}} = \sum_j p_j\varphi_j$, and the covariance is $a^2n\sum_j p_j\boldsymbol{\Sigma}_{W,j} + a^2np(1-p)\boldsymbol{\Delta}\boldsymbol{\Delta}^\mathsf{T}$ at $J = 2$	averaging $\prod_j \varphi_j^{m_j}$ over $\boldsymbol{m} \sim \text{Mult}(n, \boldsymbol{p})$ gives φ_{mix}^n ($n \leq 8$ at $J = 2$, $n \leq 5$ at $J = 3$); law of total covariance with $\operatorname{Cov}\boldsymbol{m} = n\{\operatorname{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\mathsf{T}\}$ returns the printed term; $\partial_p^2 = -2a^2n\boldsymbol{\Delta}\boldsymbol{\Delta}^\mathsf{T} \neq \boldsymbol{0}$	exact	pass	951, 1045
V7	*Maximum likelihood of the linear mixed model against DeCovarT's fit*				
V7a	The mixed model's marginal log-likelihood in $\boldsymbol{p}$ equals DeCovarT's Gaussian log-likelihood, $\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu}\boldsymbol{p}, \sum_j p_j^2\boldsymbol{\Sigma}_j)$, when $\boldsymbol{D} = \boldsymbol{0}$	$G = 1$, $J = 2$: integrating x_1, x_2 out of the joint density gives $\mathcal{N}(y;\, m(p),\, p^2s_1 + (1-p)^2s_2 + d)$, whose limit at $d = 0$ is DeCovarT's density; $G = J = 2$, general 2×2 $\boldsymbol{\Sigma}_j$: the cumulant generating function gives $\boldsymbol{V} = \sum_j p_j^2\boldsymbol{\Sigma}_j + \boldsymbol{D}$ and the two log-likelihoods agree at $\boldsymbol{D} = \boldsymbol{0}$	exact	pass	962, 1056
V7b	With a nugget $\boldsymbol{D} \neq \boldsymbol{0}$ the maximum-likelihood estimate of $\boldsymbol{p}$ differs from DeCovarT's in general	one gene, $\mu_1 = 0$, $\mu_2 = 1$, $\sigma_1^2 = 1$, $\sigma_2^2 = 4$, $y = 1/2$: $\hat{p} = 0.746679$ at $d = 0$ (DeCovarT) and $\hat{p} = 0.748539$ at $d = 1$ (exact real roots of the score). Deviation: the note's claim holds only at $\boldsymbol{D} = \boldsymbol{0}$, which is DeCovarT's assumption (A5)	exact	**flag**	963, 1057

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ID	Claim	Detail	Kind	Status	Line
V7c	Restricted maximum likelihood does not separate $\boldsymbol{p}$ from the variance: the error contrasts' covariance $\boldsymbol{K}^\mathsf{T}\boldsymbol{V}(\boldsymbol{p})\boldsymbol{K}$ depends on $\boldsymbol{p}$	$G=3$, $J=2$, $\boldsymbol{K}^\mathsf{T}\boldsymbol{\Delta}=\boldsymbol{0}$, diagonal $\boldsymbol{\Sigma}_1=\operatorname{diag}(a_g)$, $\boldsymbol{\Sigma}_2=\operatorname{diag}(b_g)$; for $\boldsymbol{\Delta}\propto\boldsymbol{e}_1$, $\partial_p(\boldsymbol{K}^\mathsf{T}\boldsymbol{V}\boldsymbol{K})=\begin{bmatrix}2\,(a_2p+b_2p-b_2) & 0\\ 0 & 2\,(a_3p+b_3p-b_3)\end{bmatrix}$, not identically $\boldsymbol{0}$. Deviation: the note's 'REML is then an option' assumed a variance free of the fixed effect	exact	**flag**	965, 1059
V8 *Hellinger and 2-Wasserstein distances against the Mahalanobis distance*					
V8a	Common $\boldsymbol{\Sigma}$: $H^2=1-\exp(-\frac{1}{8}\boldsymbol{\Delta}^\mathsf{T}\boldsymbol{\Sigma}^{-1}\boldsymbol{\Delta})$, increasing in the Mahalanobis distance	completing the square: $\sqrt{f_1f_2}=e^{-M/8}\,\mathcal{N}(\boldsymbol{x};\bar{\boldsymbol{\mu}},\boldsymbol{\Sigma})$ (polynomial identity, general 2×2 $\boldsymbol{\Sigma}$); one-dimensional integral $=e^{-\frac{(a-b)^2}{8s}}$; $\mathrm{d}H^2/\mathrm{d}M=\frac{e^{-\frac{M}{8}}}{8}>0$	exact	pass	271
V8b	Common $\boldsymbol{\Sigma}$: $W_2=\|\boldsymbol{\Delta}\|$, which does not depend on $\boldsymbol{\Sigma}$	for any coupling, $\mathbb{E}\|\boldsymbol{X}-\boldsymbol{Y}\|^2=\|\boldsymbol{\Delta}\|^2+\operatorname{tr}\operatorname{Cov}(\boldsymbol{X}-\boldsymbol{Y})\geq\|\boldsymbol{\Delta}\|^2$ (exact identity, symbolic cross-covariance); the translation $\boldsymbol{Y}=\boldsymbol{X}-\boldsymbol{\Delta}$ attains it	exact	pass	273
V8c	Counter-example: W_2 and the Mahalanobis distance can rank two pairs of type laws in opposite orders	$\boldsymbol{\Sigma}=\begin{bmatrix}1&0\\0&100\end{bmatrix}$; $\boldsymbol{\Delta}_A=(1,0)$: $W_2=1$, $M=1$; $\boldsymbol{\Delta}_B=(0,3)$: $W_2=3$, $M=\frac{9}{100}$. W_2 prefers B; M and H prefer A	exact	pass	273
V9 *Depth and Poisson sampling in the within-type covariance*					
V9a	eq:depth: $\operatorname{Cov}\boldsymbol{x}=\mathbb{E}(s^2)\boldsymbol{\Sigma}^\lambda+\operatorname{Var}(s)\,\boldsymbol{\mu}^\lambda\boldsymbol{\mu}^{\lambda\mathsf{T}}+\mathbb{E}(s)\operatorname{diag}(\boldsymbol{\mu}^\lambda)$, for Poisson counts given $(\boldsymbol{\lambda},s)$ and $s\perp\boldsymbol{\lambda}$	Poisson moments from the pmf; conditional moments given $(\boldsymbol{\lambda},s)$; tower rule with only the first two moments of s and $\boldsymbol{\lambda}$ ($G=2$); the within term is $\mathbb{E}(s)\operatorname{diag}(\boldsymbol{\mu}^\lambda)$ and the between term $\operatorname{Cov}(s\boldsymbol{\lambda})$, as in A.5	exact	pass	731, 1108
V9b	Fang & Pachter's eq. 5: for $\Sigma^\lambda_{ab}=0$, $\operatorname{Cov}(x_a,x_b)/(\mathbb{E}x_a\,\mathbb{E}x_b)=\operatorname{Var}(s)/(\mathbb{E}s)^2$, which also equals the normalised variance of a gene with $\Sigma^\lambda_{gg}=0$	normalised covariance $=\frac{-(\mathbb{E}s)^2+\mathbb{E}(s^2)}{(\mathbb{E}s)^2}$ and normalised variance $=\frac{-(\mathbb{E}s)^2+\mathbb{E}(s^2)}{(\mathbb{E}s)^2}$, with $\mathbb{E}s$ free; both are invariant to $s\mapsto cs$. Deviation: the note's 'up to the normalisation $\mathbb{E}s=1$' is not needed	exact	**flag**	668, 1112
V10 *Rank of the empirical covariance and existence of the Gaussian MLE*					
V10a	With the mean estimated, $\operatorname{rank}\boldsymbol{V}=\operatorname{rank}\boldsymbol{X}_c\leq\min(C-1,G)$, with equality for data in general position; $\boldsymbol{V}\succ0$ needs $C\geq G+1$	$G=3$, symbolic data: $\det\boldsymbol{V}\equiv0$ at $C=3$; at $C=4$, $\det\boldsymbol{V}$ is a non-zero polynomial (value $\frac{1723969}{1024}$ at a rational point); exact ranks at rational points for $C=2,\dots,6$: 1, 2, 3, 3, 3	exact	pass	299, 320, 648
V10b	An exact linear or affine relation between genes, or a constant gene, lowers the rank below $\min(C-1,G)$	$G=3$, $x_3=2x_1-x_2+5$: $\boldsymbol{V}(2,-1,-1)^\mathsf{T}=\boldsymbol{0}$; rank (affine, constant gene) $=(2,2)$ at $C=4$ and $(2,2)$ at $C=10$, against $\min(C-1,G)=3$	exact	pass	302, 308, 320, 648
V10c	If $\boldsymbol{V}$ is singular, the Gaussian log-likelihood is unbounded above: the MLE does not exist	$C=G=3$; $\boldsymbol{\Sigma}_\varepsilon=\boldsymbol{V}+\varepsilon\boldsymbol{u}\boldsymbol{u}^\mathsf{T}$, $\boldsymbol{V}\boldsymbol{u}=\boldsymbol{0}$: $\operatorname{tr}(\boldsymbol{V}\boldsymbol{\Sigma}_\varepsilon^{-1})=2$ for every ε, $\det\boldsymbol{\Sigma}_\varepsilon\propto\varepsilon$, and $\ell(\boldsymbol{\Sigma}_\varepsilon)\to\infty$ as $\varepsilon\to0^+$	exact	pass	304, 320
V10d	Zwiernik et al. (2017, pp. 1270, 1278): with the sample mean, $S_n\succ0$ almost surely "when $n\geqslant p$"; the condition is $n\geqslant p+1$	$n=p=3$: $\det S_n\equiv0$ and rank $2=p-1$ (V10a); their own $nS_n\sim W_p(n-1,\boldsymbol{\Sigma})$ (p. 1274) is singular at $n=p$	exact	**flag**	365
V11 *Linear shrinkage is positive definite*					
V11a	2×2: $(1-\rho)\boldsymbol{V}+\rho\boldsymbol{T}\succ0$ for $\boldsymbol{V}\succeq0$, $\boldsymbol{T}\succ0$, $\rho\in(0,1]$	$\boldsymbol{V}=\boldsymbol{L}\boldsymbol{L}^\mathsf{T}$, $\boldsymbol{T}=\boldsymbol{M}\boldsymbol{M}^\mathsf{T}$: $\det=(1-\rho)^2\det\boldsymbol{V}+\rho^2\det\boldsymbol{T}+\rho(1-\rho)\|\operatorname{adj}(\boldsymbol{L})\boldsymbol{M}\|_F^2\geq\rho^2\det\boldsymbol{T}>0$ (polynomial identity); trace $(1-\rho)\operatorname{tr}\boldsymbol{V}+\rho\operatorname{tr}\boldsymbol{T}>0$	exact	pass	322
V11b	Any G: $\lambda_{\min}\{(1-\rho)\boldsymbol{V}+\rho\boldsymbol{T}\}\geq\rho\,\lambda_{\min}(\boldsymbol{T})$	$\boldsymbol{x}^\mathsf{T}\boldsymbol{V}\boldsymbol{x}=\|\boldsymbol{X}_c\boldsymbol{x}\|^2/C\geq0$ (identity with symbolic data, $G=3$, $C=2$, $\operatorname{rank}\boldsymbol{V}=1$), so $\boldsymbol{x}^\mathsf{T}\hat{\boldsymbol{\Sigma}}\boldsymbol{x}\geq\rho\,\boldsymbol{x}^\mathsf{T}\boldsymbol{T}\boldsymbol{x}\geq\rho\,\lambda_{\min}(\boldsymbol{T})\|\boldsymbol{x}\|^2$	exact	pass	322, 343, 648
V11c	Numeric check of V11b at $G=40$, $C=2$, with the diagonal and the scaled-identity targets	mpmath, 40 digits, seed 20261030; $\operatorname{rank}\boldsymbol{V}=1$; for $\rho\in\{0.01,0.5,1\}$ and both targets, $\lambda_{\min}(\hat{\boldsymbol{\Sigma}})\geq\rho\,\lambda_{\min}(\boldsymbol{T})>0$ (smallest gap 0 to 30 digits, attained at $\rho=1$)	numeric	pass	322
V12 *Quantile universal threshold against Bonferroni, FWER and false edges*					
V12a	QUT $\leq$ Bonferroni quantile for any dependence; under independence the QUT is the Sidak quantile, with per-test level $1-(1-\alpha)^{1/m}\geq\alpha/m$	union bound; $\frac{\mathrm{d}}{\mathrm{d}\alpha}\{(1-\alpha/m)^m-(1-\alpha)\}=1-\left(-\frac{\alpha}{m}+1\right)^{m-1}\geq0$, zero at $\alpha=0$; $\lim_{m\to\infty}m\{1-(1-\alpha)^{1/m}\}=-\log(1-\alpha)$, so the per-test levels differ by the factor $-\log(1-\alpha)/\alpha$: 1.026 at $\alpha=0.05$, 1.386 at $\alpha=0.5$	exact	pass	576, 578

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ID	Claim	Detail	Kind	Status	Line
V12b	For independent Gaussian scores the QUT and the Bonferroni quantile agree asymptotically: ratio $\to 1$, difference $\to 0$	mpmath, 50 digits; $q_{\mathrm{QUT}}/q_{\mathrm{B}}$ at $m = 100$: 0.998053 ($\alpha = 0.05$), 0.962322 ($\alpha = 0.5$); at $m = 10^6$: 0.999167, 0.987458; the ratio increases in m and the gap $q_{\mathrm{B}} - q_{\mathrm{QUT}}$ shrinks as $\log\{-\log(1-\alpha)/\alpha\}/q_{\mathrm{B}}$ (within 10\,\% at $m = 10^6$)	numeric	pass	576
V12c	With k true and $m - k$ null regulators, the QUT controls the per-target FWER at $\le \alpha$; expected false edges $\to -\log(1-\alpha)$ per target	$\mathrm{FWER} = 1 - (1-\alpha)^{(m-k)/m}$; $\lim_{m\to\infty} \mathbb{E}V\,(k = \sqrt{m}) = -\log(1-\alpha)$; $m = 100$, $k = 10$: $\mathbb{E}V = 0.6217$, share $\mathbb{E}V/(\mathbb{E}V + k) = 0.0585$ at $\alpha = 0.5$; 0.04615 and 0.00459 at $\alpha = 0.05$; share q needs per-test level $t = \frac{kq}{(1-q)(-k+m)}$ (solved exactly)	exact	pass	571, 580, 582, 584, 654
V12d	The paper's $\lambda = 0.05N_c/\sqrt{N_g}$ grows as N_c, the null score as $\sqrt{N_c}$	$\lambda/(\sigma\sqrt{N_c}\,q) = \frac{\sqrt{N_c}}{20\sqrt{N_g}q\sigma} \to \infty$ as $N_c \to \infty$; $\lambda\sqrt{N_g} = 0.05N_c$	exact	pass	552, 558
V13 *Raw or Poisson-denoised within-type covariance in the bulk*					
V13a	Bulk covariance under cell-depth sampling then scaling, $a^2\sum_j m_j(\boldsymbol{\Sigma}_j^{\mathrm{bio}} + \mathrm{diag}\,\boldsymbol{\mu}_j)$, against bulk-depth sampling, $a^2\sum_j m_j\boldsymbol{\Sigma}_j^{\mathrm{bio}} + a\,\mathrm{diag}(\sum_j m_j\boldsymbol{\mu}_j)$; binomial thinning equals bulk-depth sampling	Poisson and binomial MGFs summed from the pmf; generic third-order CGF per type, $G = J = 2$; Hessian of the pooled CGF at 0; thinning and bulk-depth sampling have the same CGF	exact	pass	750
V13b	With $m_j = np_j$: $\mathrm{Cov}\,\boldsymbol{y} = \phi\,\boldsymbol{S}^{\mathrm{bio}}(\boldsymbol{p}) + c\,\mathrm{diag}(\tilde{\boldsymbol{\mu}}\boldsymbol{p}) = \phi\{\boldsymbol{S}^{\mathrm{bio}}(\boldsymbol{p}) + a^{-1}\,\mathrm{diag}(\tilde{\boldsymbol{\mu}}\boldsymbol{p})\}$; the raw plug-in $\phi\,\boldsymbol{S}^{\mathrm{raw}}(\boldsymbol{p})$ is exact iff $a = 1$; the denoised one only as $a \to \infty$; sampling exceeds biology for gene g iff $\tilde{\mu}_g(\boldsymbol{p}) > aS_{gg}^{\mathrm{bio}}(\boldsymbol{p})$	raw minus bulk-depth covariance: $an(a-1)(\tilde{\boldsymbol{\mu}}\boldsymbol{p})_g$ on the diagonal, 0 off it; its root in a is 1; limit of $\mathrm{Cov}\,\boldsymbol{y}/\phi - \boldsymbol{S}^{\mathrm{bio}}$ as $a \to \infty$ is 0	exact	pass	754, 809
V13c	Toy example: $\tilde{\boldsymbol{\mu}} = (10, 1)$, $\boldsymbol{\Sigma}^{\mathrm{bio}} = \begin{pmatrix} 40 & 2 \\ 2 & 1 \end{pmatrix}$; $\mathrm{Cov}\,\boldsymbol{y}/\phi$ at $a = \frac{1}{4}, 1, 4$; gene-2 variance ratios raw/true and denoised/true; crossover depths $\frac{1}{4}$ and 1	$a = \frac{1}{4}$: $\begin{pmatrix} 80 & 2 \\ 2 & 5 \end{pmatrix}$, correlation $\frac{1}{10}$, ratios $\frac{2}{5}$, $\frac{1}{5}$; $a = 1$: raw, correlation $\frac{1}{5}$, ratios 1, $\frac{1}{2}$; $a = 4$: $\begin{pmatrix} 42.5 & 2 \\ 2 & 1.25 \end{pmatrix}$, correlation 0.2744, ratios $\frac{8}{5}$, $\frac{4}{5}$	exact	pass	754
V14 *Denoised within-type covariance: technical size factor and finite-sample bias*					
V14a	$1 + \nu = (1 + \nu^{\mathrm{bio}})(1 + \nu^{\mathrm{tech}})$ for $s = s^{\mathrm{bio}}s^{\mathrm{tech}}$, independent factors of mean 1	$\mathbb{E}s = 1$ and $\mathrm{Var}\,s = \nu^{\mathrm{bio}}\nu^{\mathrm{tech}} + \nu^{\mathrm{bio}} + \nu^{\mathrm{tech}}$ from the product of moments	exact	pass	767
V14b	eq-count-covariance, and the population version of eq-denoised-within equals the target $\mathrm{Cov}(\boldsymbol{B}s^{\mathrm{bio}}\boldsymbol{\lambda}) = \boldsymbol{B}[(1 + \nu^{\mathrm{bio}})\boldsymbol{\Sigma}^{\lambda} + \nu^{\mathrm{bio}}\boldsymbol{\mu}\boldsymbol{\mu}^{\mathsf{T}}]\boldsymbol{B}$	Poisson moments from the pmf; tower rule over $(s, \boldsymbol{\lambda})$, $G = 2$, gene-wise capture $\boldsymbol{B}$; $\boldsymbol{V} \to \mathrm{Cov}\,\boldsymbol{x}$, $\bar{\boldsymbol{x}} \to \mathbb{E}\boldsymbol{x}$ gives the target exactly	exact	pass	769
V14c	eq-denoised-within as first printed is biased: $\mathbb{E}\hat{\boldsymbol{\Sigma}} - \mathrm{target} = -\kappa\,\mathrm{Cov}\,\boldsymbol{x}/C_j$, $\kappa = \nu^{\mathrm{tech}}/(1 + \nu^{\mathrm{tech}})$; replacing $\bar{\boldsymbol{x}}\bar{\boldsymbol{x}}^{\mathsf{T}}$ by $\bar{\boldsymbol{x}}\bar{\boldsymbol{x}}^{\mathsf{T}} - \boldsymbol{V}/C_j$ makes it unbiased	$\mathbb{E}\bar{\boldsymbol{x}}\bar{\boldsymbol{x}}^{\mathsf{T}} = \mathbb{E}\boldsymbol{x}\,\mathbb{E}\boldsymbol{x}^{\mathsf{T}} + \mathrm{Cov}\,\boldsymbol{x}/C$ and $\mathbb{E}\boldsymbol{V} = \mathrm{Cov}\,\boldsymbol{x}$ from the double sum over cells, symbolic C. Deviation: the printed form is consistent, biased at order $1/C_j$; corrected in 05-1	exact	**flag**	769
V14d	Plugging the total extrinsic noise ν (a size factor carrying s^{bio} and s^{tech}) for ν^{tech} returns $\boldsymbol{B}\boldsymbol{\Sigma}^{\lambda}\boldsymbol{B}$ and drops $\nu^{\mathrm{bio}}\boldsymbol{B}(\boldsymbol{\Sigma}^{\lambda} + \boldsymbol{\mu}\boldsymbol{\mu}^{\mathsf{T}})\boldsymbol{B}$	population estimator with $\kappa = \nu/(1 + \nu)$, ν from V14a; difference from the target factorised	exact	pass	772, 807
V14e	Monte Carlo of V14c: gamma size factors ($\nu^{\mathrm{bio}} = 0.3$, $\nu^{\mathrm{tech}} = 0.5$), log-normal $\boldsymbol{\lambda}$ with correlated genes, $C_j = 10$ cells	numpy seed 20261031, 2×10^5 replicates; largest $\lvert z\rvert$: printed form against target minus bias 1.40, corrected form against target 1.41 (tolerance 4.5); printed form against target 11.7	numeric	pass	769
V15 *Gaussian transport map (Bisque's gene-wise map as its diagonal case)*					
V15a	eq-bures-map: $\boldsymbol{A}$ symmetric positive definite, $\boldsymbol{A}\boldsymbol{\Sigma}_1\boldsymbol{A} = \boldsymbol{\Sigma}_2$, and the displacement cost equals $\lVert\boldsymbol{m}_1 - \boldsymbol{m}_2\rVert^2 + \mathrm{tr}\{\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2 - 2(\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2})^{1/2}\}$	$\boldsymbol{\Sigma}_1 = \mathrm{diag}(1, 4)$, $\boldsymbol{\Sigma}_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$; closed-form 2×2 square root checked by squaring; both costs equal exactly (≈ 10.7712)	exact	pass	qmd
V15b	Same identities for $G = 4$	mpmath at 40 digits, 20 random SPD pairs (seed 20261031); largest residual 6.4×10^{-40}, tolerance 10^{-30}; all eigenvalues of $\boldsymbol{A}$ positive	numeric	pass	qmd

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ID	Claim	Detail	Kind	Status	Line
V15c	Diagonal covariances: $\boldsymbol{A} = \mathrm{diag}(\sigma_{2g}/\sigma_{1g})$, Bisque's gene-wise map; one dimension: $W_2^2 = (m_1 - m_2)^2 + (\sigma_1 - \sigma_2)^2$	entrywise square roots of diagonal matrices, symbolic	exact	pass	qmd
V16	*Bulk sampling diagonal against the variance scale: identifiability and profile*				
V16a	Covariance alone, given $\boldsymbol{p}$: $(\phi, k) \mapsto \phi\, \boldsymbol{S}^{\mathrm{bio}}(\boldsymbol{p}) + k\, \mathrm{diag}(\tilde{\boldsymbol{\mu}}\boldsymbol{p})$ is identifiable iff $\boldsymbol{S}^{\mathrm{bio}}(\boldsymbol{p})$ is not a multiple of $\mathrm{diag}(\tilde{\boldsymbol{\mu}}\boldsymbol{p})$	Jacobian $[\mathrm{vec}\,\boldsymbol{S}, \mathrm{vec}\,\boldsymbol{D}]$: all 2×2 minors vanish iff $\boldsymbol{S} = \kappa \boldsymbol{D}$ ($G = 2, 3$, generic symmetric $\boldsymbol{S}$); $\boldsymbol{S} = \boldsymbol{D}/2$, $\boldsymbol{m} = (2, 3)$: $(\phi, k) = (4, 6)$ and $(8, 4)$ give $\mathrm{diag}(16, 24)$; Fisher determinant 0 there, $\frac{7}{678976}$ for a non-proportional $\boldsymbol{S}$	exact	pass	qmd
V16b	With the mean $c\,\tilde{\boldsymbol{\mu}}\boldsymbol{p}$ of the article and $\mathrm{Cov}\,\boldsymbol{y} = \phi\, \boldsymbol{S}^{\mathrm{bio}}(\boldsymbol{p}) + c\, \mathrm{diag}(\tilde{\boldsymbol{\mu}}\boldsymbol{p})$, (c, ϕ), and so (ϕ, a), is identified whenever $\boldsymbol{S}^{\mathrm{bio}}(\boldsymbol{p}) \neq \boldsymbol{0}$, also when it is a multiple of the diagonal	Fisher information of (c, ϕ) from the expected Hessian, $G = 2$: $\boldsymbol{I} = \boldsymbol{I}_{\mathrm{cov}} + \mathrm{diag}(\boldsymbol{m}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{m}, 0)$, $\det \boldsymbol{I} = (\boldsymbol{m}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{m}) I_{\phi\phi} + \det \boldsymbol{I}_{\mathrm{cov}}$; at $\boldsymbol{S} = \kappa \boldsymbol{D}$: $\det \boldsymbol{I}_{\mathrm{cov}} = 0$, $\det \boldsymbol{I} = \frac{\kappa^2 (m_1 + m_2)}{(c + \kappa\phi)^3}$; $\partial(c, \phi)/\partial(\phi, a)$ has determinant ϕ/a^2	exact	pass	qmd
V16c	Cost: with $\phi = ac$ and $\boldsymbol{M} = a\boldsymbol{S}^{\mathrm{bio}}(\boldsymbol{p}) + \mathrm{diag}(\tilde{\boldsymbol{\mu}}\boldsymbol{p})$, $\hat{c}$ for fixed $(\boldsymbol{p}, a)$ is the positive root of $Cc^2 + Gc - A = 0$; a needs a one-dimensional numerical profile; without the diagonal, $\hat{c}$ is the GLS scale and $\hat{\phi} = Q/G$	$2c^2\, \partial_c \ell = A - Gc - Cc^2$, $A = \boldsymbol{y}^\top \boldsymbol{M}^{-1} \boldsymbol{y}$, $C = \boldsymbol{m}^\top \boldsymbol{M}^{-1} \boldsymbol{m}$ ($G = 2$, generic $\boldsymbol{M}$); one positive root; $\partial_c^2 \ell = -G/(2c^2) - C/c < 0$ there	exact	pass	qmd
V17	*Presence gate: zero-variance and single-detection genes*				
V17a	A gene with zero counts in every cell of type j gives a zero row and column of $\boldsymbol{V}_j$, $\det \boldsymbol{V}_j = 0$, and an undefined (0/0) correlation with every other gene	Symbolic counts, $C = 4$, $G = 3$: $v_{1h} = 0$ for all h, $\det \boldsymbol{V}_j = 0$, rank ≤ 2; $v_{12} = 0$ and $v_{11} v_{22} = 0$	exact	pass	qmd
V17b	A gene detected in one cell c only has correlation $\tilde{h}_c / (\sqrt{1 - 1/C_j}\, \|\tilde{\boldsymbol{h}}\|)$ with gene h; two genes detected only in the same cell have correlation 1 and a singular covariance	Symbolic counts, $C = 4$: formula holds; counts $(a, 0, 0, 0)$, $(b, 0, 0, 0)$: $r = 1$, $\det = 0$	exact	pass	qmd
V17c	If $\delta_{\min} C_j \geq 2$, a gene that passes $\delta_{gj} \geq \delta_{\min}$ is non-zero in at least two cells of type j, so the cases of V17a and V17b are excluded	Linear certificate $n_z - 2 = C_j(n_z/C_j - \delta_{\min}) + (\delta_{\min} C_j - 2)$, both terms ≥ 0 (n_z non-zero cells)	exact	pass	qmd
V18	*D-, E- and A-optimal gene panels from the Fisher information for $\boldsymbol{p}$*				
V18a	With a common within-type covariance, $\boldsymbol{S}_W(\boldsymbol{p}) = \boldsymbol{\Sigma}_W$ and the information for the free coordinates $\boldsymbol{\theta}$ ($\boldsymbol{p} = \boldsymbol{p}_0 + \boldsymbol{K}\boldsymbol{\theta}$) is $n_i \boldsymbol{K}^\mathsf{T} \boldsymbol{M}^\mathsf{T} \boldsymbol{\Sigma}_W^{-1} \boldsymbol{M} \boldsymbol{K}$; for $J = 2$ it is $n_i \boldsymbol{\Delta}^\mathsf{T} \boldsymbol{\Sigma}_W^{-1} \boldsymbol{\Delta}$	Expected negative Hessian of the Gaussian log-likelihood, $G = 2$, $J = 3$, generic $\boldsymbol{M}$ and $\boldsymbol{\Sigma}_W$; Hessian free of $\boldsymbol{y}$; $n_i = c^2/\phi$	exact	pass	qmd
V18b	With $\boldsymbol{\Sigma}_W = \boldsymbol{I}$ the information is $n_i \boldsymbol{A}^\mathsf{T} \boldsymbol{A}$, $\boldsymbol{A} = \boldsymbol{M}\boldsymbol{K}$: its determinant is the squared Gram volume of the contrasted mean columns and $\kappa(\boldsymbol{I}) = \kappa(\boldsymbol{A})^2$	Lagrange identity $\det \boldsymbol{A}^\mathsf{T} \boldsymbol{A} = \|\boldsymbol{a}_1\|^2 \|\boldsymbol{a}_2\|^2 - (\boldsymbol{a}_1^\mathsf{T} \boldsymbol{a}_2)^2$ for generic 3×2 $\boldsymbol{A}$; $\kappa(\boldsymbol{A}) = 1.954$ (A), 2.136 (B)	exact	pass	qmd
V18c	$\mathrm{Cov}\,\hat{\boldsymbol{p}} = \boldsymbol{K} \boldsymbol{I}_{\boldsymbol{K}}^{-1} \boldsymbol{K}^\mathsf{T}$ does not depend on the basis $\boldsymbol{K}$ and $\det \boldsymbol{I}_{\boldsymbol{K}\boldsymbol{B}} = (\det \boldsymbol{B})^2 \det \boldsymbol{I}_{\boldsymbol{K}}$; E and A computed in the raw coordinates (p_1, p_2) are basis-dependent	Generic $\boldsymbol{K}$, $\boldsymbol{B}$, $\boldsymbol{I}_{\boldsymbol{K}}$; counter-example: $\lambda_{\min}$ of two panels 4.50 vs 6.84 in (p_1, p_2), 3.05 vs 2.30 with orthonormal $\boldsymbol{K}$	exact	pass	qmd
V18d	Example ($J = 3$, $G = 3$): with $\boldsymbol{\Sigma}_W = \boldsymbol{I}$ panel A beats panel B on D, E and A; with correlation 4/5 between genes 1 and 3 of panel A, D still prefers A while E and A prefer B	Orthonormal $\boldsymbol{K}$, $n_i = 1$. A, $\rho = 0$: D $\frac{116}{3}$, E $\frac{23}{3} - \frac{\sqrt{181}}{3}$, A $\frac{23}{58}$; B: D $\frac{104}{3}$, E $\frac{23}{3} - \frac{\sqrt{217}}{3}$, A $\frac{23}{52}$; A, $\rho = 4/5$: D 60, E $\frac{47}{3} - \frac{\sqrt{1669}}{3}$, A $\frac{47}{90}$	exact	pass	qmd
V19	*Explicit tail bound for RNA-Sieve's range filter*				
V19a	The bulk mean $b_g = \sum_k \alpha_k m_{gk}$ lies in $[\min_k m_{gk}, \max_k m_{gk}]$, and $\delta(\hat{b}_g, [\min_k \tilde{m}_{gk}, \max_k \tilde{m}_{gk}]) \geq t$ implies $\|\hat{b}_g - b_g\| \geq t/2$ or $\max_k \|\tilde{m}_{gk} - m_{gk}\| \geq t/2$	Linear certificates: $t - (L - \hat{b})$ and $t - (\hat{b} - U)$ written as sums of terms positive under the hypotheses, checked by expansion	exact	pass	qmd
V19b	With expression in $[0, B]$, Hoeffding and the union bound give $P(\delta \geq t) \leq 2e^{-nt^2/(2B^2)} + 2\sum_k e^{-c_k t^2/(2B^2)}$	Hoeffding $2e^{-2Nu^2/B^2}$ at $u = t/2$ equals $2e^{-Nt^2/(2B^2)}$; Hoeffding itself quoted	exact	pass	qmd

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ID	Claim	Detail	Kind	Status	Line
V19c	Monte Carlo: the empirical tail of δ stays below the bound of V19b	Binomial counts, $B = 2$, $K = 3$, $n = c_k = 100$, $\boldsymbol{\alpha} = (0.98, 0.01, 0.01)$, seed 20261101, 2×10^4 replicates, 40 values of t; tail 0.215 at $t = 0.06$; first bound below 1 at $t = 0.42$: 0.882 against 0.0000 (conservative)	numeric	pass	qmd
V20	*Negative-binomial counts are sub-exponential: moments, Chernoff and Bernstein range bounds*				
V20a	For $X \sim \mathrm{NB}(r, \pi)$, $M(\lambda) = \{(1-\pi)/(1-\pi e^\lambda)\}^r$, finite exactly for $\lambda < -\log\pi$, with mean $\mu = r\pi/(1-\pi)$ and variance $\mu + \mu^2/r$: the NB is sub-exponential	Coefficients of $(1-z)^{-r}$ to z^9 equal $(r)_x/x!$ (symbolic r); $M'(0)$, $M''(0) - M'(0)^2$ simplify; ratio test limit πe^λ	exact	pass	qmd
V20b	Moment growth of Prop. 2.8.1(ii) for the centred NB: $(\mathbb{E}\lvert X-\mu\rvert^p)^{1/p}/p$ stays bounded in p	$r \in \{0.5, 2, 10\}$, $\pi \in \{0.3, 0.6, 0.9\}$, $p = 1, \ldots, 60$, log-space sums in float64; largest ratio 23.74 at $(r, \pi) = (10.0, 0.9)$; ratios for $p > 30$ never exceed those for $p \le 30$	numeric	pass	qmd
V20c	Cramér–Chernoff rate of the NB, $I(a) = a\log\frac{a(\mu+r)}{\mu(a+r)} + r\log\frac{\mu+r}{a+r}$, with $I(\mu) = 0$, $I(0) = r\log(1+\mu/r)$ and the Poisson limit; Monte Carlo tails of the mean and of the range distance δ stay below the Chernoff bounds	Stationary point and closed form symbolic; MC seed 20261101: mean of $n = 50$ $\mathrm{NB}(\mu = 3, r = 2)$, 2×10^5 replicates, at $u = 1$ tail 0.0080 vs bound 0.0584; range filter $K = 3$, $\mu_k = (1, 3, 6)$, $r = 2$, $\boldsymbol{\alpha} = (0.98, 0.01, 0.01)$, $n = c_k = 100$, 2×10^4 replicates; first bound below 1 at $t = 1.3$: 0.950 vs 0.0000	numeric	pass	qmd
V21	*Unico: confounded variance components and spectral entropy*				
V21a	If $\mathbf{1}^\top \boldsymbol{w} = 1$, then $\boldsymbol{w}^\top(\boldsymbol{\Sigma}_g + c\mathbf{1}\mathbf{1}^\top)\boldsymbol{w} = \boldsymbol{w}^\top\boldsymbol{\Sigma}_g\boldsymbol{w} + c$: a constant shift of Unico's cross-type covariance is confounded with τ_g^2	Symbolic 4×4 symmetric $\boldsymbol{\Sigma}$, $w_4 = 1 - w_1 - w_2 - w_3$; difference expands to 0	exact	pass	qmd
V21b	Spectral entropy of a $k \times k$ correlation matrix: $0 \le H \le \log k$, $H = \log k$ at $\boldsymbol{R} = \mathbf{I}$, $H = 0$ at $\boldsymbol{R} = \mathbf{1}\mathbf{1}^\top$, and the Rényi-2 bound $H \ge \log[k/\{1 + (k-1)\overline{\rho^2}\}]$	$k \in \{2, 3, 5, 10, 50\}$, 200 random correlation matrices each, seed 20261101; smallest H 0.002, largest $H/\log k$ 1.000; tolerance 10^{-12}	numeric	pass	qmd
V22	*CSNet's shared-gamma negative binomials and the factor covariance*				
V22a	CSNet's Algorithm 1: $\mathbb{E}x = sT\phi$, $\mathrm{Var}\,x = s^2T\phi^2 + sT\phi$, $\mathrm{Cov}(x_g, x_{g'}) = s^2\phi^2 T_{gg'}$, $\mathrm{Cor} = (T_{gg'}/T)\, s\phi/(1+s\phi)$; $\mathrm{Var}\,x = \sigma^2$ gives $s\phi = \sqrt{\sigma^2/T + 1/4} - 1/2$	Law of total (co)variance with a shared $\mathrm{Gamma}(T_{gg'}, 1)$ component; positive root of $Tu^2 + Tu = \sigma^2$	exact	pass	qmd
V22b	CSNet constants and simulation: $b = 0.77$ and $\mathbb{E}x \approx 678$ at $s = 6 \times 10^7$, $\sigma^2 = e^8$, $T = 200$; Monte Carlo of Algorithm 1 matches the moments; an AR-type block violates $\sum_{g'} T_{gg'} \le T$	$b = 0.7724$, $\mathbb{E}x = 678.6$; MC MA(1), $\rho = 0.5$, $G = 6$, $s = 1$, $\phi = 2$, 2×10^5 draws, seed 20261101: mean 400.0 (exact 400), variance 1202 (exact 1200), lag-one correlation 0.334 (exact 0.333), lag two -0.000; AR-type row sum 14.7	numeric	pass	qmd
V22c	Woodbury: $(\boldsymbol{\Psi} + \boldsymbol{B}\boldsymbol{B}^\top)^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}\boldsymbol{B}(\mathbf{I}_q + \boldsymbol{B}^\top\boldsymbol{\Psi}^{-1}\boldsymbol{B})^{-1}\boldsymbol{B}^\top\boldsymbol{\Psi}^{-1}$	$G = 4$, $q = 2$, symbolic loadings and diagonal $\boldsymbol{\Psi}$; every entry of the difference cancels	exact	pass	qmd
V23	*Multivariate Fisher information for $\boldsymbol{p}$ and the Bhattacharyya overlap, $J = 2$*				
V23a	Multivariate Fisher information for p, $J = 2$: $I(p) = n_i\boldsymbol{\Delta}^\top\boldsymbol{S}(p)^{-1}\boldsymbol{\Delta} + \frac{1}{2}\,\mathrm{tr}[\{\boldsymbol{S}(p)^{-1}(\boldsymbol{\Sigma}_{W,1} - \boldsymbol{\Sigma}_{W,2})\}^2]$	$-\mathbb{E}\,\partial^2\ell/\partial p^2$ from the Gaussian log-likelihood, $G = 2$, rational covariances, symbolic p, c, ϕ with $n_i = c^2/\phi$	exact	pass	1082
V23b	Bhattacharyya distance of two Gaussian laws: $D_B = \frac{1}{8}\boldsymbol{\Delta}^\top\bar{\boldsymbol{\Sigma}}^{-1}\boldsymbol{\Delta} + \frac{1}{2}\log\{\det\bar{\boldsymbol{\Sigma}}/(\det\boldsymbol{\Sigma}_1 \det\boldsymbol{\Sigma}_2)^{1/2}\}$; equal covariances give $\boldsymbol{\Delta}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\Delta}/8$	$-\log\int\sqrt{f_1 f_2}$ by quadrature ($G = 1$, three cases, 30 digits) and on a 1201^2 grid ($G = 2$); largest difference 7.6e-14	numeric	pass	1097
V24	*Lyapunov covariance, conditional precision and the triangular factor of a DAG*				
V24a	Symmetric $\boldsymbol{A}$ and $\boldsymbol{Q} = q\mathbf{I}$: $\boldsymbol{\Sigma} = -\frac{q}{2}\boldsymbol{A}^{-1}$ solves the Lyapunov equation, so $\boldsymbol{\Omega} = -\frac{2}{q}\boldsymbol{A}$ and $\omega_{gg} = 2a_g/q$	$G = 3$, symbolic symmetric $\boldsymbol{A}$; residual of the Lyapunov equation and $\boldsymbol{\Omega} + \frac{2}{q}\boldsymbol{A}$ simplify to zero	exact	pass	875
V24b	Linear SEM on a directed acyclic graph, parents first: $\boldsymbol{\Omega} = (\mathbf{I} - \boldsymbol{B})^\top\boldsymbol{D}^{-1}(\mathbf{I} - \boldsymbol{B})$ and $\omega_{gg} = 1/d_g + \sum_{g\to h} b_{hg}^2/d_h$	$G = 3$, symbolic strictly lower triangular $\boldsymbol{B}$ and diagonal $\boldsymbol{D}$; inverse of $(\mathbf{I} - \boldsymbol{B})^{-1}\boldsymbol{D}(\mathbf{I} - \boldsymbol{B})^{-\top}$	exact	pass	897, 929
V24c	$\omega_{gg} = 1/\mathrm{Var}(x_g \mid \boldsymbol{x}_{-g}) = 1/\{\sigma_{gg}(1 - R_g^2)\} > 0$	$G = 3$, covariance with one symbolic entry t; Schur complement against the multiple correlation, every gene	exact	pass	881, 928

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ID	Claim	Detail	Kind	Status	Line
V25	*Intrinsic noise, bulk scales and naive intervals*				
V25a	Linear-noise approximation of one gene: Fano factor 1 for birth–death, $1+b$ for geometric bursts of mean size b	Fixed point, noise intensity q and the one-gene Lyapunov equation $-2a\sigma+q=0$; geometric moments from the generating function	exact	pass	qmd
V25b	With $c_i=a_in_i$ and $\phi_i=a_i^2n_i$: $a_i=\phi_i/c_i$ and $n_i=c_i^2/\phi_i$	Direct substitution	exact	pass	not cited
V25c	Two genes, correlation ρ, same signature entry: OLS and GLS share the variance $(1+\rho)/(2m^2)$, the naive independence variance is $1/(2m^2)$	Sandwich, GLS and naive variances computed symbolically	exact	pass	not cited
V26	*Dynamic and mixed Gaussian Bayesian networks*				
V26a	Linear Gaussian DBN: the stationary covariance solves $\boldsymbol{\Sigma}=\boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^\top+\boldsymbol{Q}$, $(\mathbf{I}-\boldsymbol{A}\otimes\boldsymbol{A})\operatorname{vec}\boldsymbol{\Sigma}=\operatorname{vec}\boldsymbol{Q}$; a positive two-gene loop between slices gives $\sigma_{12}\neq 0$, a symmetric negative loop gives $\sigma_{12}=0$ but a non-zero lag-one cross-covariance $(\boldsymbol{A}\boldsymbol{\Sigma})_{12}$	Symbolic 2×2 $\boldsymbol{A}$ with persistence $1/2$ and cross edges $\pm 1/4$, $\boldsymbol{Q}=\mathbf{I}$; residuals vanish; positive loop $\sigma_{12}=64/105$; negative loop $\sigma_{12}=0$, $(\boldsymbol{A}\boldsymbol{\Sigma})_{12}=-4/11$; spectral radii 0.750, 0.559	exact	pass	1301, 1304
V26b	Without couplings, $\boldsymbol{A}'=\boldsymbol{A}\boldsymbol{M}\boldsymbol{U}\boldsymbol{M}^{-1}$ ($\boldsymbol{M}=\boldsymbol{\Sigma}_t^{1/2}$, $\boldsymbol{U}$ orthogonal fixing $\boldsymbol{M}^{-1}\boldsymbol{\mu}_t$) gives the same $\boldsymbol{\mu}_{t+1}$ and $\boldsymbol{\Sigma}_{t+1}$ as $\boldsymbol{A}$	$G=3$, seed 20261102: mean error 2.2e-16, covariance error 1.8e-15, $\max\lvert\boldsymbol{A}'-\boldsymbol{A}\rvert=0.45$	numeric	pass	1313, 1458
V26c	Mixture covariance $\sum_k\pi_k\boldsymbol{\Sigma}_k+\sum_k\pi_k(\boldsymbol{\mu}_k-\bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_k-\bar{\boldsymbol{\mu}})^\top$: diagonal classes give the off-diagonal $\pi_1\pi_2(\mu_{11}-\mu_{21})(\mu_{12}-\mu_{22})$	Second moments of a two-class mixture, $G=2$, symbolic	exact	pass	1358
V26d	$X\to Y$ and $Y\to X$ with matched parameters have the same observational law but differ under $\operatorname{do}(X=x_0)$: $Y\sim\mathcal{N}(bx_0,s_e^2)$ against $\mathcal{N}(0,b^2s_x^2+s_e^2)$	Covariances of the two SEMs equal symbolically; interventional means and variances differ for $b\neq 0$	exact	pass	1416
V26e	v-structure $X\to Z\leftarrow Y$: $\Sigma_{xy}=0$ but $J_{xy}=1/d\neq 0$ (moralisation in the precision)	$Z=X+Y+e$, $e\sim\mathcal{N}(0,d)$; covariance and precision computed symbolically	exact	pass	1272

C Gene selection in second-generation deconvolution methods

Table 5 summarises the review behind Section 3, in the order of the tiers of the feature-selection vignette, whose comparison table holds the same methods and facts. Tiers (i) and (ii) are filters, tier (iii) is the weighting in the likelihood, and cross-platform consistency is a perspective there, since it needs a bulk matched to the reference. Pages are PDF pages of the local files (main text unless 'S' for the supplement); a dash means not reported in the parts we extracted.

Table 5: Gene selection in deconvolution methods by tier, with CardamomOT for comparison. Abbreviations are listed in Section 1.

Method	Cell QC (tier i)	Gene filter (tier i)	Marker rule (tier ii)	Weighting (tier iii)	Cross-platform (perspective)
RNA-Sieve (Erdmann-Pham et al. 2021)	optional: library size or pairwise distance ≥ 3 MAD from the type median (S p. 3)	non-zero in $\geq 20\,\%$ of the cells of ≥ 1 type (S p. 3)	none	Gaussian composite likelihood from the CLT (eq. 10, p. 10)	range bound and MAD outliers on four statistics (eqs 12–13, p. 11)
DECALS (Cai et al. 2024)	–	CIBERSORTx S-mode candidate genes (p. 13)	$\cap$ markers of Mathys et al. (Fisher's exact test, FDR): 159 genes (p. 13; S pp. 18–19)	constrained OLS; GLS biased in practice (pp. 6, 9)	S-mode batch correction (p. 13)
MEAD (Xie et al. 2026)	–	–	none; removing genes gains nothing unless noise differs (p. 17)	WLS from the reference variance (pp. 17–18); sandwich (p. 13)	gene-wise ratios with mean 1 (pp. 5, 8)
ReDeconv (Lu et al. 2025)	–	–	t-test $p < 0.05$ against every other type, no FDR; FC ≥ 2; smallest CV, about 100 per type (p. 14)	Gaussian, composition-dependent variance (eq. 5, p. 14)	length normalisation of the bulk only (p. 2)
MuSiC (Wang et al. 2019)	–	all shared genes (p. 7)	none; small within-cluster variance in the tree-guided stage (pp. 3, 8)	W-NNLS, weights from cross-subject variance (eq. 10, p. 7)	optional gene- and subject-specific intercept (p. 7)
deMixSC (Guo et al. 2023)	< 500 UMIs, < 200 genes, $> 50\,\%$ mitochondrial (p. 11)	detected in $< 5\,\%$ of cells removed (p. 11)	none	weighted W-NNLS, iterated weights (p. 12)	paired t-test bulk vs pseudo-bulk, BH; 5,000 genes down-weighted by $a = 1000$ (p. 12)
BayesPrism (Chu et al. 2022)	types with < 20 cells excluded in one benchmark (S p. 15)	ribosomal, mitochondrial, X and Y genes (S pp. 14, 19)	optional scran markers, t-tests, up-regulated only (S pp. 14–15)	Bayesian multinomial (S pp. 3–6)	bulk-specific profiles; log-normal fold-change prior (S pp. 5–6)
DWLS (Tsoucas et al. 2019)	–	–	MAST hurdle, FDR < 0.01, $\log_2$FC > 0.5; 50–200 per type; smallest condition number (p. 7)	dampened WLS (pp. 2, 7)	none
SCDC (Dong et al. 2021)	cells with a questionable type or low quality removed (p. 9)	genes with zero counts in every sample (p. 9)	none	W-NNLS, maximal-variance weights clipped at the 15th/85th percentiles (p. 10)	ENSEMBLE over references (p. 3)
RCTD (Cable et al. 2022)	spots ≥ 100 UMIs (p. 9)	mean above 0.0625 counts per 500 (p. 7)	log-fold change ≥ 0.5 against the other types, no test (p. 7)	Poisson–lognormal GLMM (p. 6)	gene-wise platform random effect (pp. 3, 6)
Cardamo-mOT 2.0.0	paper: –; code: per-cell depth factor, no cell removed	paper: gene sets of earlier studies (p. 32); code: no mitochondrial, ribosomal or non-coding genes	code: W_1 and entropy terminals, round robin of per-type Wilcoxon tests, NB separability, Steiner closure	NB-mixture likelihood, ℓ_1 towards a prior network (eq. 10, p. 9)	not applicable (no bulk)
DeCovarT	authors' QC; MAD rules optional	presence gate in every type	omnibus NB or quasi-likelihood F test as an FDR gate, effect-size rank	GLS with $\hat{\boldsymbol{\Sigma}}_{W,j}$	none: one reference specimen

D Purely statistical approaches to directed networks

This appendix sketches a statistical counterpart to CardamomOT: a directed, signed network with a likelihood, but without a mechanistic bursting model. The candidate is a mixture of linear Gaussian dynamic Bayesian networks (DBNs). Nothing here is implemented; the aim is to name the tools and the research tracks, with their conditions. Book pages refer to the printed pages of Koller and Friedman (2009) (KF below).

D.1 Gaussian Bayesian networks

- **Linear Gaussian conditional laws.** Each gene given its parents is $x_g \mid \boldsymbol{x}_{\mathrm{pa}(g)} \sim \mathcal{N}(\beta_{g0} + \boldsymbol{\beta}_g^{\mathsf{T}} \boldsymbol{x}_{\mathrm{pa}(g)}, \sigma_g^2)$ (KF Def. 5.14, p. 187). A directed acyclic graph (DAG) of such laws defines a joint Gaussian, and conversely every Gaussian is a linear Gaussian network for any ordering of the genes (KF Thms 7.3–7.5, pp. 251–253); the network is 'not unique', so its direction is not read from the joint law. In matrix form this is the SEM of Equation (9).
- **What the precision encodes.** A zero of $\boldsymbol{J} = \boldsymbol{\Sigma}^{-1}$ is a conditional independence given all other genes (KF Thm 7.2, p. 251): the edges of the precision are those of the moral graph, not of the DAG. Two independent parents of one gene have $\Sigma_{xy} = 0$ but $J_{xy} \neq 0$ (KF Ex. 7.5, p. 254) [V26e]. A sparse DAG can therefore have a dense precision, and a graphical lasso returns the moral graph at best.
- **Fitting.** Gaussian laws form an exponential family, and the maximum-likelihood fit of a network is the moment-matching projection $\prod_g P(x_g \mid \boldsymbol{x}_{\mathrm{pa}(g)})$ (KF Thms 8.6–8.7, pp. 278–281): one regression per gene on its parents.

D.2 Dynamic Bayesian networks: self-regulation and feedback through time

- **Two-slice template.** A DBN assumes a Markov transition, stationary in time, and describes it by a two-slice network over $(\boldsymbol{x}^{(t)}, \boldsymbol{x}^{(t+1)})$, unrolled over any horizon (KF Defs 6.1–6.4, pp. 201–204). Edges inside a slice must form a DAG; edges between slices include persistence edges $x_g^{(t)} \to x_g^{(t+1)}$ (p. 204). The linear Gaussian case is

$$\boldsymbol{x}^{(t+1)} = \boldsymbol{A}\boldsymbol{x}^{(t)} + \boldsymbol{e}^{(t+1)}, \qquad \boldsymbol{e}^{(t+1)} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \tag{12}$$

  (KF eqs 6.3–6.4, pp. 211–212), with $A_{gh} \neq 0$ an edge $h^{(t)} \to g^{(t+1)}$.
- **Loops without cycles.** The diagonal A_{gg} is a persistence edge: the lag-one effect of a gene on itself, the dynamic counterpart of self-regulation. Two edges of opposite direction between slices, $h^{(t)} \to g^{(t+1)}$ and $g^{(t)} \to h^{(t+1)}$, form a feedback loop. The unrolled network stays acyclic; the loops live in the influence graph, which KF define as cyclic ('a persistence arc $X \to X'$ induces a self-cycle', Def. 15.2, p. 658). Time also orients these edges: 'In DBNs, we cannot reverse an edge $A \to A'$' (p. 837), so the direction that a static network cannot identify (Appendix D.1) is fixed between slices.
- **Stationary covariance.** A stable (12) has a stationary covariance that solves the discrete Lyapunov (Stein) equation, the counterpart of the continuous equation of Section 5,

$$\boldsymbol{\Sigma} = \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^{\mathsf{T}} + \boldsymbol{Q}, \qquad (\boldsymbol{I} - \boldsymbol{A} \otimes \boldsymbol{A}) \operatorname{vec} \boldsymbol{\Sigma} = \operatorname{vec} \boldsymbol{Q} \quad \text{[V26a]}. \tag{13}$$

  A symmetric negative loop between two genes (one activates, the other represses, with equal rates) gives $\sigma_{12} = 0$ but a non-zero lag-one cross-covariance $(\boldsymbol{A}\boldsymbol{\Sigma})_{12}$ [V26a]: the loop is invisible in one snapshot and visible across time (our remark).
- **Inference over time.** The Kalman filter propagates $\boldsymbol{\mu}^{(t+1)} = \boldsymbol{A}\boldsymbol{\mu}^{(t)}$ and $\boldsymbol{\Sigma}^{(t+1)} = \boldsymbol{A}\boldsymbol{\Sigma}^{(t)}\boldsymbol{A}^{\mathsf{T}} + \boldsymbol{Q}$ (KF eqs 15.4–15.5, pp. 676–677). Over time, every pair of state variables becomes dependent (entanglement, Thm 15.1, p. 658), and with switching regimes the exact belief state is a Gaussian mixture whose size grows exponentially in t (pp. 683–685).
- **The snapshot problem.** Sequencing destroys the cell, so no cell is observed at t and at $t+1$. If only the moments of consecutive snapshots are used, $\boldsymbol{A}$ and $\boldsymbol{A}' = \boldsymbol{A}\boldsymbol{M}\boldsymbol{U}\boldsymbol{M}^{-1}$, with $\boldsymbol{M} = \boldsymbol{\Sigma}_t^{1/2}$

and $\boldsymbol{U}$ an orthogonal matrix that fixes $\boldsymbol{M}^{-1}\boldsymbol{\mu}_t$, give the same $\boldsymbol{\mu}_{t+1}$ and $\boldsymbol{\Sigma}_{t+1}$ [V26b]: two snapshots do not identify the network (our remark). Couplings between time points are needed: optimal transport, as in CardamomOT; a pseudotime; RNA velocity or metabolic labelling, which measure a time derivative within one cell (La Manno et al. 2018); or interventions (Appendix D.5).

D.3 Structure learning

- **Constraint-based** methods (PC type) test conditional independences and return an I-equivalence class, drawn as a partially directed graph. They need bounded in-degree, faithfulness and a reliable test, and are 'sensitive to failures in individual independence tests' (KF pp. 785–790).
- **Score-based** methods maximise a penalised likelihood. The likelihood alone favours the complete graph (Prop. 18.1, pp. 792–794); BIC, $\ell(\hat{\theta}_{\mathcal{G}}) - \frac{1}{2}\log M \dim \mathcal{G}$, is consistent (Thm 18.2, pp. 802–803) and decomposes over genes (Def. 18.2, p. 805). The Bayesian score with a BDe prior is score-equivalent (Thm 18.4, p. 807), and BGe, its Gaussian counterpart with a normal–Wishart prior, is cited only (p. 840). Finding the best graph is NP-hard for in-degree two or more (Thm 18.5, p. 811); search uses greedy edge moves, GES, which is consistent under faithfulness (pp. 821–822), or MCMC over structures or orders (pp. 829–832).
- **A known order helps.** With the order of the genes known, the best parents of each gene are chosen independently (Prop. 18.4, p. 810), polynomial for bounded in-degree. In a DBN, time supplies that order for the edges between slices; with hundreds of genes, a node-wise sparse regression (lasso on $\boldsymbol{A}_{g\cdot}$) is the scalable version of the same idea (research track).

D.4 Mixtures of (dynamic) Bayesian networks

- **Model.** Each cell c carries a latent regime z_c (a type, a basin), with its own transition (Figure 2):

$$z_c \sim \mathrm{Cat}(\boldsymbol{\pi}), \qquad \boldsymbol{x}_c^{(t+1)} \mid \boldsymbol{x}_c^{(t)}, z_c = k \sim \mathcal{N}\big(\boldsymbol{A}_k \boldsymbol{x}_c^{(t)}, \boldsymbol{Q}_k\big). \tag{14}$$

- **Estimation.** The expectation–maximisation (EM) algorithm increases the likelihood at every step and converges to a stationary point only (KF Thms 19.3–19.4, pp. 874–875); with Gaussian classes its E-step is a soft classification (pp. 875–877). Structural EM also searches the graph of each class: the score gain on the completed data bounds the true gain from below (Thm 19.10), but the result is not proved to be a local maximum (Algorithm 19.3, p. 922). With missing data, BIC 'tends to underfit badly' (p. 915).
- **Mixtures mimic feedback.** If two classes orient one edge in opposite directions, the network induced by the mixture is cyclic, 'nevertheless … it specifies a coherent distribution' (KF Box 5.B, multinets, pp. 170–171).
- **Pooling creates edges.** A mixture of classes with independent genes is correlated: $\mathrm{Cov}(\boldsymbol{x}) = \sum_k \pi_k \boldsymbol{\Sigma}_k + \sum_k \pi_k (\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}})^{\mathsf{T}}$ [V26c]. A network fitted on pooled types, time points or basins shows edges that no class has; this is why the basin label matters in CardamomOT.
- **Research tracks.** None of the verified methods is both dynamic and built for counts; the switching linear dynamical system is the closest dynamic form (KF pp. 683–685).
  - *clustNet* clusters samples by a mixture of DAGs, each component learnt with the MCMC samplers of *BiDAG*; the data must be binary or categorical (Bayer et al. 2025; Kuipers et al. 2022).
  - VaMSL learns a mixture of causal DAGs by variational inference, extending DiBS, with expert knowledge as a prior (Björkman et al. 2026); a covariate-dependent mixture lets covariates set the membership probabilities (Marchant et al. 2025).
  - Identifiability: what can be learnt from data generated by a mixture of DAGs (Saeed et al. 2020), and which edges interventions identify in such a mixture (Varıcı et al. 2024).
  - pyAgrum's `BNMixture` is not a latent-class model: it is a list of discrete networks learnt separately, with weights set by the user, or a Bayesian bootstrap of one network (`BootstrapMixture`); its use here is to measure the stability of each arc (pyAgrum developers 2026).

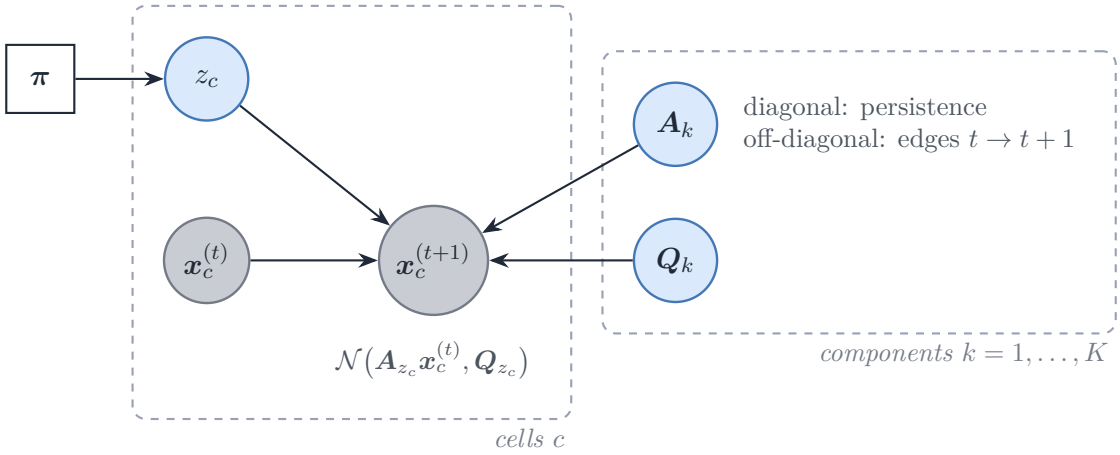


Figure 2: A mixture of linear Gaussian DBNs, Equation (14). Each cell c has a latent regime z_c with prior weights $\boldsymbol{\pi}$; given $z_c = k$, its profile at $t+1$ is Gaussian around $\boldsymbol{A}_k \boldsymbol{x}_c^{(t)}$. The diagonal of $\boldsymbol{A}_k$ holds the persistence edges, its off-diagonal the edges between slices. With snapshot data, the pair $(\boldsymbol{x}_c^{(t)}, \boldsymbol{x}_c^{(t+1)})$ comes from a coupling between time points, not from one cell. Blue: latent; grey: observed; square: fixed.

D.5 Interventions and causality

- **Intervention queries.** A causal model is a DAG whose do-operator cuts the edges into the intervened genes (KF Def. 21.1, pp. 1014–1015). Observational data identify at best the I-equivalence class (pp. 1041–1043); $X \to Y$ and $Y \to X$ can share one Gaussian law and differ under $\mathrm{do}(X = x_0)$ [V26d].
- **Interventional likelihood.** With the genes Z fixed in a sample, the likelihood keeps the conditional laws of the other genes only, $\prod_{g \notin Z} P(x_g \mid \boldsymbol{x}_{\mathrm{pa}(g)})$ (KF p. 1044), and 'we can now distinguish between I-equivalent models'. With latent confounders the target becomes a partial ancestral graph (pp. 1048–1050).
- **A worked method.** Rau et al. (2013) fit one linear Gaussian causal network to observational and knock-out data together: each sample k has its own intervention set J_k, whose equations drop out of the likelihood (their eq. 2, p. 4); Metropolis–Hastings runs over gene orderings with a Mallows proposal, and the parameters of each ordering have a closed form. In simulation (10 genes) the AUROC of total effects rose from 0.749 with observational data alone to 0.948 with one knock-out per gene (their Table 1, p. 6). The method has no sparsity, costs $O(p^6)$, and 'feedback loops … cannot be modeled' (pp. 10–11).
- **Relevance to single cells.** A Perturb-seq design gives each cell its own target set J_c, which is exactly this likelihood. Three gaps remain: CRISPR interference lowers expression rather than fixing it (a soft intervention), counts need a count emission, and cell types need the mixture of Appendix D.4. KF note the same limit for signalling networks: 'the assumption of acyclicity … does not hold for cellular networks' (Box 21.D, pp. 1046–1047), which the dynamic model of Appendix D.2 relaxes.

D.6 Undirected models, briefly

A GGM estimates the moral graph, not the regulatory one (Appendix D.1), but it is the better-understood route in high dimension: joint graphical lassos with group or cooperative penalties share structure across conditions (Chiquet et al. 2011; Chiquet et al. 2012), the PLN network adds the count layer (Chiquet et al. 2021), and the conditions for support recovery are those of Section 3.3 (Chiquet 2015). Mixtures of GGMs estimate the clusters and their networks together: SCAN (Hao et al. 2018), multiple networks in a Gaussian mixture (Gao et al. 2016), AJGM, with a network shared across clusters (Yang et al. 2025), and BLGGM for zero-inflated single-cell data (Wu and Luo 2022); PLNmodels offers mixtures and networks of the PLN model as separate model variants. A survey of joint estimation is Tsai et al. (2022).

D.7 What it would give CardamomOT, and open problems

- **Gains:** signed, directed edges with a likelihood, hence held-out scores to set the penalty (Section 7), intervals, likelihood-ratio tests, and a direct use of perturbation data.
- **Open problems:** a count emission (Poisson–log-normal rather than Gaussian), sparsity with hundreds of genes, the identifiability of a mixture of networks (label switching, classes that differ by their means rather than their graphs), and couplings between snapshots, without which the transition is not identified [V26b].